



Top 54 CAT Quant Based DI Questions With Video Solutions

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Questions

Instructions

Two students, Amiya and Ramya are the only candidates in an election for the position of class representative. Students will vote based on the intensity level of Amiya's and Ramya's campaigns and the type of campaigns they run. Each campaign is said to have a level of 1 if it is a staid campaign and a level of 2 if it is a vigorous campaign. Campaigns can be of two types, they can either focus on issues, or on attacking the other candidate.

If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2)\%$, that is, 80% of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and another 10% who would have otherwise voted for Amiya, will not vote at all.
- If Ramya runs a campaign attacking Amiya and Amiya runs a campaign focusing on issues, then 20% of the students who would have otherwise voted for Ramya will vote for Amiya, and another 5% who would have otherwise voted for Ramya, will not vote at all.
- If both run campaigns attacking each other, then 10% of the students who would have otherwise voted for them had they run campaigns focusing on issues, will not vote at all.

Question 1

If Amiya runs a campaign focusing on issues, then what is the maximum percentage of votes that she can get?

- A 48%
- B 44%
- C 40%
- D 36%

A	R	TV
1	1	40%
1	2	60%

Handwritten calculations: 40%, 60%, 20%, 4%, 20%, 20%.

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Question 2

If both of them run staid campaigns attacking the other, then what percentage of students will vote in the election?

- A 40%
- B 64%
- C 60%
- D 36%

Two students, Amiya and Ramya are the only candidates in an election for the position of class representative. Students will vote based on the intensity level of Amiya's and Ramya's campaigns and the type of campaigns they run. Each campaign is said to have a level of 1 if it is a staid campaign and a level of 2 if it is a vigorous campaign. Campaigns can be of two types, they can either focus on issues, or on attacking the other candidate.

If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2) = 80\%$ of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and



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Question 3

If Ramya runs a campaign attacking Amiya, then what is the minimum percentage of votes that she is guaranteed to get?

- A 12%
- B 15%
- C 30%
- D 18%

Two students, Amiya and Ramya are the only candidates in an election for the position of class representative. Students will vote based on the intensity level of Amiya's and Ramya's campaigns and the type of campaigns they run. Each campaign is said to have a level of 1 if it is a staid campaign and a level of 2 if it is a vigorous campaign. Campaigns can be of two types, they can either focus on issues, or on attacking the other candidate.

If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2) = 80\%$ of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and



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Instructions

Directions for the following five questions: Answer the following questions based on the information given below:

Abdul, Bikram and Chetan are three professional traders who trade in shares of a company XYZ Ltd. Abdul follows the strategy of buying at the opening of the day at 10 am and selling the whole lot at the close of the day at 3 pm. Bikram follows the strategy of buying at hourly intervals: 10 am, 11 am, 12 noon, 1 pm and 2 pm, and selling the whole lot at the close of the day. Further, he buys an equal number of shares in each purchase. Chetan follows a similar pattern as Bikram but his strategy is somewhat different. Chetan's total investment amount is divided equally among his purchases. The profit or loss made by each investor is the difference between the sale value at the close of the day less the investment in purchase. The "return" for each investor is defined as the ratio of the profit or loss to the investment amount expressed as a percentage.

Question 4

One day, two other traders, Dane and Emily joined Abdul, Bikram and Chetan for trading in the shares of XYZ Ltd. Dane followed a strategy of buying equal numbers of shares at 10 am, 11 am and 12 noon, and selling the same numbers at 1 pm, 2 pm and 3 pm. Emily, on the other hand, followed the strategy of buying shares using all her money at 10 am and selling all of them at 12 noon and again buying the shares for all the money at 1 pm and again selling all of them at the close of the day at 3 pm. At the close of the day the following was observed.

- Abdul lost money in the transactions.
- Both Dane and Emily made profits.
- There was an increase in share price during the closing hour compared to the price at 2 pm.
- Share price at 12 noon was lower than the opening price

Share price was at its highest at

- A 10 am
- B 11 am
- C 12 noon
- D 1 pm
- E cannot be determined

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Instructions

Question Numbers: (55 to 58)

In a square layout of site 5m ~ 5m 25 equal-sized square platforms of different heights are built. The heights (in metre) of individual platforms are as shown below:

6	1	2	4	3
9	5	3	2	8
7	8	4	6	5
3	9	5	1	2
1	7	6	3	9

Individuals (all of same height) are seated on these platforms. We say an individual A can reach individual B, if all the three following conditions are met;

- A and B are in the same row or column
- A is at a lower height than B
- If there is/are any individual(s) between A and B, such individual(s) must be at a height lower than that of A.

Thus in the table given above, consider the Individual seated at height 8 on 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

Question 5

We can find two individuals who cannot be reached by anyone in

- A the last row.
- B the fourth row.
- C the fourth column.
- D the middle column.



1	6	1	2	4	3
2	9	5	3	2	8
3	7	8	4	6	5
4	3		5	1	2
5	1			3	9

Thus in the table given above, consider the individual seated at height 8 on 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

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Instructions

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

The kids surveyed were further divided into two groups based on whether their mothers dropped out of school before completing primary education or not.. The table below gives the number of kids in different types of schools for mothers who dropped out of school before completing primary education:

	G	P	O	Total
NE	4200	500	300	5000
W	4200	1900	1200	7300
S	5100	300	300	5700
TOTAL	13500	2700	1800	18000

It is also known that:

1. In S, 60% of the surveyed kids were in G. Moreover, In S, all surveyed kids whose mothers had completed primary education were in school.
2. In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
3. The number of kids in G in NE was the same as the number of kids in G in W.

Question 6

Among the kids in W whose mothers had completed primary education, how many were not in school?

- A 300
- B 1200
- C 1050
- D 1500

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NE = 7500 W = 12500 S = 10,000

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VIDEO SOLUTION

CAT Previous Papers PDF

Instructions

	column 1	column 2	column 3
row 1			
row 2			
row 3			

	Column 1	Column 2	Column 3
Row 1	(2, 4)	(6, 8)	(1, 3)
Row 2	(3, 5)	(1, 1)	(6, 20)
Row 3	(1, 2)	(1, 2)	(2, 5)

Three pouches (each represented by a filled circle) are kept in each of the nine slots in a 3×3 grid, as shown in the figure. Every pouch has a certain number of one-rupee coins. The minimum and maximum amounts of money (in rupees) among the three pouches in each of the nine slots are given in the table. For example, we know that among the three pouches kept in the second column of the first row, the minimum amount in a pouch is Rs. 6 and the maximum amount is Rs. 8.

There are nine pouches in any of the three columns, as well as in any of the three rows. It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. It is also known that the total amount of money kept in the three pouches in the first column of the third row is Rs. 4.

Question 7

What is the number of slots for which the average amount (in rupees) of its three pouches is an integer?

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	Column 1	Column 2	Column 3
Row 1	2, 6	6, 6	1, 1
Row 2	3, 5	5, 5	1, 1
Row 3	3, 2	2, 2	0, 0

Three pouches (each represented by a filled circle) are kept in each of the nine slots in a 3×3 grid, as shown in the figure. Every pouch has a certain number of one-rupee coins. The minimum and maximum amounts of money (in rupees) among the three pouches in each of the nine slots are given in the table. For example, we know that among the three pouches kept in the second column of the first row, the minimum amount in a pouch is Rs. 6 and the maximum amount is Rs. 8.

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VIDEO SOLUTION

Instructions

There are only four brands of entry level smartphones called Azra, Bysi, Cxqi, and Dipq in a country. Details about their market share, unit selling price, and profitability (defined as the profit as a percentage of the revenue) for the year 2016 are given in the table below:

Brand	Market Share (%)	Unit Selling Price (Rs.)	Profitability (%)
Azra	40	15,000	10
Bysi	25	20,000	30
Cxqi	15	30,000	40
Dipq	20	25,000	30

In 2017, sales volume of entry level smartphones grew by 40% as compared to that in 2016. Cxqi offered a 40% discount on its unit selling price in 2017, which resulted in a 15% increase in its market share. Each of the other three brands lost 5% market share. However, the profitability of Cxqi came down to half of its value in 2016. The unit selling prices of the other three brands and their profitability values remained the same in 2017 as they were in 2016.

Question 8

The brand that had the highest revenue in 2016 is:

- A Cxqi
- B Bysi
- C Azra
- D Dipq

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	Column 1	Column 2	Column 3
Row 1	2, 6	6, 6	1, 1
Row 2	3, 5	5, 5	1, 1
Row 3	3, 2	2, 2	0, 0

Details about the market share, unit selling price, and profitability (defined as the profit as a percentage of the revenue) for the year 2016 are given in the table below:

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2017 → 140

A
B
C
D

Instructions

A country has the following types of traffic signals.

3 red lights = stop

2 red lights = turn left

1 red light = turn right

3 green lights = go at 100 km/hr speed

2 green lights = go at 40 km/hr speed

1 green light = go at 20 km/hr speed

A motorist starts at a point on a road and follows all traffic signals. His car is heading towards the north. He encounters the following signals (the time mentioned in each case below is applicable after crossing the previous signal).

Starting point - 1 green light

After half an hour, 1st signal - 2 red and 2 green lights

After 15 min, 2nd signal - 1 red light

After half an hour, 3rd signal - 1 red and 3 green lights

After 24 min, 4th signal - 2 red and 2 green lights

After 15 min, 5th signal - 3 red lights

Question 9

If at the starting point, the car was heading towards south, what would be the final position of the motorist?

- A 30 km to the east and 40 km to the south
- B 50 km to the east and 40 km to the south
- C 30 km to the west and 40 km to the south
- D 50 km to the west and 20 km to the north



Instructions

Directions for the following five questions: Answer the following questions based on the information given below:

Abdul, Bikram and Chetan are three professional traders who trade in shares of a company XYZ Ltd. Abdul follows the strategy of buying at the opening of the day at 10 am and selling the whole lot at the close of the day at 3 pm. Bikram follows the strategy of buying at hourly intervals: 10 am, 11am, 12 noon, 1 pm and 2 pm, and selling the whole lot at the close of the day. Further, he buys an equal number of shares in each purchase. Chetan follows a similar pattern as Bikram but his strategy is somewhat different. Chetan's total investment amount is divided equally among his purchases. The profit or loss made by each investor is the difference between the sale value at the close of the day less the investment in purchase. The "return" for each investor is defined as the ratio of the profit or loss to the investment amount expressed as a percentage.

Question 10

Which one of the following statements is always true?

- A Abdul will not be one with the minimum return
- B Return for Chetan will be higher than that of Bikram
- C Return for Bikram will be higher than that of Chetan
- D Return for Chetan cannot be higher than that of Abdul
- E none of the above

 [VIEW SOLUTION](#)

Question 11

On a "boom" day the share price of XYZ Ltd. keeps rising throughout the day and peaks at the close of the day. Which trader got the minimum return on that day?

- A Bikram
- B Chetan
- C Abdul
- D Abdul or Chetan
- E cannot be determined

 [VIEW SOLUTION](#)

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2 red lights = turn left

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2 green lights = go at 40 km/hr speed

1 green light = go at 20 km/hr speed

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After half an hour, 1st signal - 2 red and 2 green lights

After 15 min, 2nd signal - 1 red light

After half an hour, 3rd signal - 1 red and 3 green lights

After 24 min, 4th signal - 2 red and 2 green lights

After 15 min, 5th signal - 3 red lights

Question 12

After the starting point if the 1st signal were 1 red and 2 green lights, what would be the final position of the motorist?

- A 30 km to the west and 20 km to the south
- B 30 km to the west and 40 km to the north
- C 50 km to the east and 40 km to the north
- D Directly 30 km to the east

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Question 13

What is the position (radial distance) of the most motorist when he reaches the last signal?

- A 45 km directly north of the starting point
- B 30 km directly to the east of the starting point
- C 50 km away to the north-east of the starting point
- D 45 km away to the north-west of the starting point

[VIEW SOLUTION](#)

Instructions

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition				
Ingredient	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

Question 14

For a recuperating patient, the doctor recommended a diet containing 10% minerals and at least 30% protein. In how many different ways can we prepare this diet by mixing at least two ingredients?

- A One
- B Two
- C Three
- D Four
- E None

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (Q, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is Q: 150, P: 50, Q: 200, R: 500, S: 100.

Ingredient	Carbohydrate %	Protein %	Fat %	Minerals %
Q	45	50	0	5
P	5	50	40	5
Q	10	50	50	5
R	80	20	0	0
S	50	20	20	10

For a recuperating patient, the doctor recommended a diet containing at least 30% protein. In how many different ways can we prepare this diet by mixing at least two ingredients?

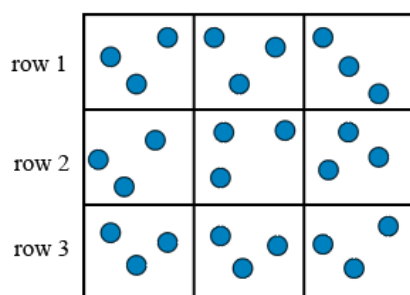
A) One
B) Two
C) Three
D) Four
E) None

Handwritten notes: Q & P, 10%, 30%, 10%, 30%, 10%, 30%

VIDEO SOLUTION

Instructions

column 1 column 2 column 3



	Column 1	Column 2	Column 3
Row 1	(2, 4)	(6, 8)	(1, 3)
Row 2	(3, 5)	(1, 1)	(6, 20)
Row 3	(1, 2)	(1, 2)	(2, 5)

Three pouches (each represented by a filled circle) are kept in each of the nine slots in a 3×3 grid, as shown in the figure. Every pouch has a certain number of one-rupee coins. The minimum and maximum amounts of money (in rupees) among the three pouches in each of the nine slots are given in the table. For example, we know that among the three pouches kept in the second column of the first row, the minimum amount in a pouch is Rs. 6 and the maximum amount is Rs. 8.

There are nine pouches in any of the three columns, as well as in any of the three rows. It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. It is also known that the total amount of money kept in the three pouches in the first column of the third row is Rs. 4.

Question 15

How many pouches contain exactly one coin?

Handwritten notes: c1, c2, c3

	Column 1	Column 2	Column 3
Row 1	2, 4	6, 8	1, 3
Row 2	3, 5	1, 1	6, 20
Row 3	1, 2	1, 2	2, 5

Handwritten notes: 2, 4, 6, 8, 1, 3, 3, 5, 1, 1, 6, 20, 1, 2, 1, 2, 2, 5

Three pouches (each represented by a filled circle) are kept in each of the nine slots in a 3×3 grid, as shown in the figure. Every pouch has a certain number of one-rupee coins. The minimum and maximum amounts of money (in rupees) among the three pouches in each of the nine slots are given in the table. For example, we know that among the three pouches kept in the second column of the first row, the minimum amount in a pouch is Rs. 6 and the maximum amount is Rs. 8.

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VIDEO SOLUTION

Instructions

The following table represents addition of two six-digit numbers given in the first and the second rows, while the sum is given in the third row. In the representation, each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 has been coded with one letter among A, B, C, D, E, F, G, H, J, K, with distinct letters representing distinct digits.

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

Question 16

Which digit does the letter A represent?

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The following table represents addition of two six-digit numbers given in the first and the second rows, while the sum is given in the third row. In the representation, each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 has been coded with one letter among A, B, C, D, E, F, G, H, J, K, with distinct letters representing distinct digits.

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

A -
B -
C -
D -
E -
F - 0
G -

H -
K -

VIDEO SOLUTION

Question 17

Which digit does the letter B represent?

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The following table represents addition of two six-digit numbers given in the first and the second rows, while the sum is given in the third row. In the representation, each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 has been coded with one letter among A, B, C, D, E, F, G, H, J, K, with distinct letters representing distinct digits.

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

A -
B -
C -
D -
E -
F - 0
G -

H -
K -

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Question 18

Which among the digits 3, 4, 6 and 7 cannot be represented by the letter D?

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The following table represents addition of two six-digit numbers given in the first and the second rows, while the sum is given in the third row. In the representation, each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 has been coded with one letter among A, B, C, D, E, F, G, H, I, K, with distinct letters representing distinct digits.

	B	B	A	A	G
	A	H	J	D	K
	A	F	G	C	E

A -
B -
C -
D -
E -
F -
G -

H -
K -

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Question 19

Which among the digits 4, 6, 7 and 8 cannot be represented by the letter G?

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The following table represents addition of two six-digit numbers given in the first and the second rows, while the sum is given in the third row. In the representation, each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 has been coded with one letter among A, B, C, D, E, F, G, H, I, K, with distinct letters representing distinct digits.

	B	B	A	A	G
	A	H	J	D	K
	A	F	G	C	E

A -
B -
C -
D -
E -
F -
G -

H -
K -

VIDEO SOLUTION

Instructions

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

A currency exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. The buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The following facts are known about the outlet on a particular day:

1. The amount of L used by the outlet to buy C equals the amount of L it received by selling C.
2. The amounts of L used by the outlet to buy A and B are in the ratio 5:3.
3. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.
4. The outlet received 88000 units of L by selling A during the day.
5. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.
6. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 51000 units of C.

Question 20

What was the base exchange rate of currency B with respect to currency L on that day ?

Q.1

A → 100k. 95k. 110k.
 B → 120k. 114k. 132k.
 C → K. 0.95k. 1.1k.
 Base Buying Selling

	Bought			
A				
B				
C				

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

A currency exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:120:132. The buying exchange rates of each of A, B, and C with respect to L are 5% below their corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The exchange rates are known at the outlet on a particular day:

1. The amount of L used by the outlet to buy C equals the amount of L it received by selling A during the day.

2. The amounts of L the outlet received from the sales of A and B are in the ratio 5:3.

3. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

4. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.

5. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 4800 units of C.

VIDEO SOLUTION

Instructions

There are only four brands of entry level smartphones called Azra, Bysi, Cxqi, and Dipq in a country. Details about their market share, unit selling price, and profitability (defined as the profit as a percentage of the revenue) for the year 2016 are given in the table below:

Brand	Market Share (%)	Unit Selling Price (Rs.)	Profitability (%)
Azra	40	15,000	10
Bysi	25	20,000	30
Cxqi	15	30,000	40
Dipq	20	25,000	30

In 2017, sales volume of entry level smartphones grew by 40% as compared to that in 2016. Cxqi offered a 40% discount on its unit selling price in 2017, which resulted in a 15% increase in its market share. Each of the other three brands lost 5% market share. However, the profitability of Cxqi came down to half of its value in 2016. The unit selling prices of the other three brands and their profitability values remained the same in 2017 as they were in 2016.

Question 21

The brand that had the highest profit in 2017 is:

- A Bysi
- B Azra
- C Cxqi
- D Dipq

Details about their market share, unit selling price, and profitability (defined as the profit as a percentage of the revenue) for the year 2016 are given in the table below:

Brand	Market Share (%)	Unit Selling Price (Rs.)	Profitability (%)
Azra	40	15,000	10
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2017 → 140

A
B
C
D



CAT FORMULAS BOOK

BY 100%ILER



Instructions

Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

- Column 1 gives the country's identity.
- Column 2 gives the country's GDP as a fraction of the GDP of Country 9.
- Column 3 gives the country's GDP per capita as a fraction of the GDP per capita of Country 10.
- Column 4 gives the country's annual GDP growth rate.
- Column 5 gives the country's annual population growth rate.

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question 22

The ratio of Country 4's GDP to Country 5's GDP in 2026 will be closest to

- A 1.195
- B 0.963
- C 1.314
- D 1.032

Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

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Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question (1)

Which one among the countries 1 through 8, has the smallest population in 2024?

A) Country 8 B) Country 3
C) Country 7 D) Country 5

Population ~ GDP

Instructions

An old woman had the following assets:

- (a) Rs. 70 lakh in bank deposits
- (b) 1 house worth Rs. 50 lakh
- (c) 3 flats, each worth Rs. 30 lakh
- (d) Certain number of gold coins, each worth Rs. 1 lakh

She wanted to distribute her assets among her three children; Neeta, Seeta and Geeta.

The house, any of the flats or any of the coins were not to be split. That is, the house went entirely to one child; a flat went to one child and similarly, a gold coin went to one child.

Question 23

Among the three, Neeta received the least amount in bank deposits, while Geeta received the highest. The value of the assets was distributed equally among the children, as were the gold coins.

How many flats did Neeta receive?

cracku

An old woman had the following assets:

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Question (1)

Among the three, Neeta received the least amount in bank deposits, while Geeta received the highest. The value of the assets was distributed equally among the children, as were the gold coins.

How much did Seeta receive in bank deposits (in lakhs of rupees)?

N 70 S 70 G 70

70 - BD
50 - House
3x30 - Flats
X - Gold Coins

210

VIDEO SOLUTION

Question 24

Among the three, Neeta received the least amount in bank deposits, while Geeta received the highest. The value of the assets was distributed equally among the children, as were the gold coins.

How much did Seeta receive in bank deposits (in lakhs of rupees)?

- A 30
- B 40
- C 20
- D 10

cracku

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How much did Seeta receive in bank deposits (in lakhs of rupees)?

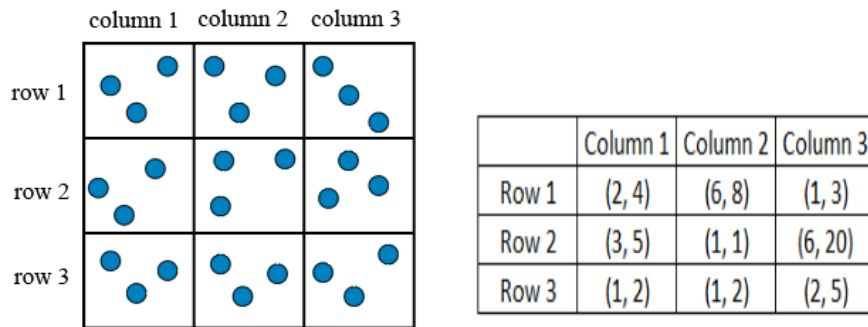
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70 - BD
50 - House
3x30 - Flats
X - Gold Coins

210

Top-500 Free CAT Questions (With Solutions)

Instructions



Three pouches (each represented by a filled circle) are kept in each of the nine slots in a 3×3 grid, as shown in the figure. Every pouch has a certain number of one-rupee coins. The minimum and maximum amounts of money (in rupees) among the three pouches in each of the nine slots are given in the table. For example, we know that among the three pouches kept in the second column of the first row, the minimum amount in a pouch is Rs. 6 and the maximum amount is Rs. 8.

There are nine pouches in any of the three columns, as well as in any of the three rows. It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. It is also known that the total amount of money kept in the three pouches in the first column of the third row is Rs. 4.

Question 25

What is the total amount of money (in rupees) in the three pouches kept in the first column of the second row?

The video solution shows a hand-drawn diagram of the 3x3 grid and a table with the minimum and maximum amounts of money in rupees for each slot. The table is identical to the one in the figure above. The video also shows a person writing the solution on a piece of paper.

Instructions

Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

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Country 8	0.07	0.41	1.2%	0.71%

Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question 26

Which one among the countries 1 through 8, has the smallest population in 2024?

- A Country 8
- B Country 3
- C Country 7
- D Country 5

Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

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Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question (1)

Which one among the countries 1 through 8, has the smallest population in 2024?

A) Country 8 B) Country 3
C) Country 7 D) Country 5

Population ~ GDP

VIDEO SOLUTION

Question 27

For how many countries among Country 1 through Country 8 will the GDP per capita in 2027 be lower than that in 2024?

Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

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Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question (1)

Which one among the countries 1 through 8, has the smallest population in 2024?

A) Country 8 B) Country 3
C) Country 7 D) Country 5

Population ~ GDP

Free CAT Study Material

Instructions

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

A currency exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. The buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The following facts are known about the outlet on a particular day:

1. The amount of L used by the outlet to buy C equals the amount of L it received by selling C.
2. The amounts of L used by the outlet to buy A and B are in the ratio 5:3.
3. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.
4. The outlet received 88000 units of L by selling A during the day.
5. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.
6. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 51000 units of C.

Question 28

How many units of currency C did the outlet sell on that day?

- A 22000
- B 19000
- C 6000
- D 3000

Handwritten solution for Question 28:

Base exchange rates: A → 100k, B → 120k, C → 1k. Buying rates: A → 95k, B → 114k, C → 0.95k. Selling rates: A → 110k, B → 132k, C → 1.1k.

	Bought			
A				
B				
C				

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

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Instructions

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

The kids surveyed were further divided into two groups based on whether their mothers dropped out of school before completing primary education or not.. The table below gives the number of kids in different types of schools for mothers who dropped out of school before completing primary education:

	G	P	O	Total
NE	4200	500	300	5000
W	4200	1900	1200	7300
S	5100	300	300	5700
TOTAL	13500	2700	1800	18000

It is also known that:

1. In S, 60% of the surveyed kids were in G. Moreover, In S, all surveyed kids whose mothers had completed primary education were in school.
2. In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
3. The number of kids in G in NE was the same as the number of kids in G in W.

Question 29

What percentage of kids from S were studying in P?

- A 37%
- B 6%
- C 79%
- D 56%



NE = 7500 W = 12500 S = 10,000

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

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NE	4200	500	300	5000
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 VIDEO SOLUTION

Instructions

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

Question 30

If the ATM could serve only 10 customers with a stock of fifty 500 rupee notes and a sufficient number of notes of other denominations, what is the maximum number of customers among these 10 who could have given 500 rupee notes as their preferences?

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

In how many different ways can the ATM serve a customer who gives 500 rupee notes as her preference?

500	200	100
50	1	1
10	2	3
9	1	5
	0	

cracku

VIDEO SOLUTION

Free CAT Preparation Video Lectures

Question 31

What is the number of 500 rupee notes required to serve 50 customers with 500 rupee notes as their preferences and another 50 customers with 100 rupee notes as their preferences, if the total number of notes to be dispensed is the smallest possible?

- A 900
- B 800
- C 750
- D 1400

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

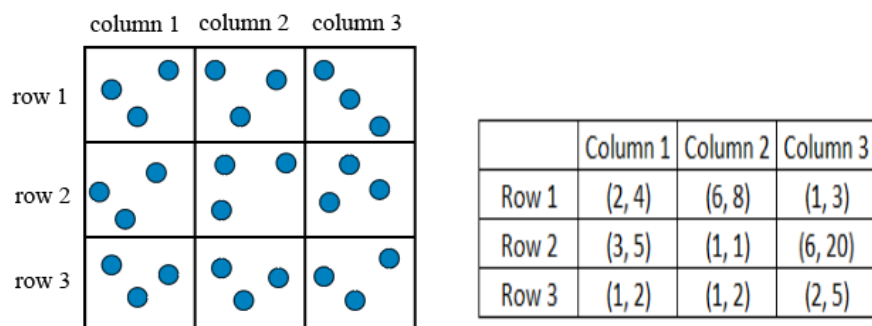
In how many different ways can the ATM serve a customer who gives 500 rupee notes as her preference?

500	200	100
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10	2	3
9	1	5
	0	

cracku

VIDEO SOLUTION

Instructions



Three pouches (each represented by a filled circle) are kept in each of the nine slots in a 3×3 grid, as shown in the figure. Every pouch has a certain number of one-rupee coins. The minimum and maximum amounts of money (in rupees) among the three pouches in each of the nine slots are given in the table. For example, we know that among the three pouches kept in the second column of the first row, the minimum amount in a pouch is Rs. 6 and the maximum amount is Rs. 8.

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Question 32

The number of slots for which the total amount in its three pouches strictly exceeds Rs. 10 is

The video solution shows a hand-drawn diagram of the 3x3 grid and a table with the minimum and maximum amounts of money in rupees for each row and column. The table is identical to the one in the figure above.

VIDEO SOLUTION

Instructions

Two students, Amiya and Ramya are the only candidates in an election for the position of class representative. Students will vote based on the intensity level of Amiya's and Ramya's campaigns and the type of campaigns they run. Each campaign is said to have a level of 1 if it is a staid campaign and a level of 2 if it is a vigorous campaign. Campaigns can be of two types, they can either focus on issues, or on attacking the other candidate.

If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2)\%$, that is, 80% of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and another 10% who would have otherwise voted for Amiya, will not vote at all.
- If Ramya runs a campaign attacking Amiya and Amiya runs a campaign focusing on issues, then 20% of the

students who would have otherwise voted for Ramya will vote for Amiya, and another 5% who would have otherwise voted for Ramya, will not vote at all.

- If both run campaigns attacking each other, then 10% of the students who would have otherwise voted for them had they run campaigns focusing on issues, will not vote at all.

Question 33

What is the minimum percentage of students who will vote in the election?

- A 32%
- B 40%
- C 38%
- D 36%

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A	R	TV	A	R
1	1	40%	20%	20%
1	2	60%	20%	4%

VIDEO SOLUTION

Daily Free CAT Practice Tests

Instructions

Directions for the following five questions: Answer the following questions based on the information given below:

Abdul, Bikram and Chetan are three professional traders who trade in shares of a company XYZ Ltd. Abdul follows the strategy of buying at the opening of the day at 10 am and selling the whole lot at the close of the day at 3 pm. Bikram follows the strategy of buying at hourly intervals: 10 am, 11am, 12 noon, 1 pm and 2 pm, and selling the whole lot at the close of the day. Further, he buys an equal number of shares in each purchase. Chetan follows a similar pattern as Bikram but his strategy is somewhat different. Chetan's total investment amount is divided equally among his purchases. The profit or loss made by each investor is the difference between the sale value at the close of the day less the investment in purchase. The "return" for each investor is defined as the ratio of the profit or loss to the investment amount expressed as a percentage.

Question 34

On a day of fluctuating market prices, the share price of XYZ Ltd. ends with a gain, i.e., it is higher at the close of the day compared to the opening value. Which trader got the maximum return on that day?

- A Bikram
- B Chetan
- C Abdul
- D Bikram or Chetan

E cannot be determined

[VIEW SOLUTION](#)

Instructions

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

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2. In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
3. The number of kids in G in NE was the same as the number of kids in G in W.

Question 35

In a follow up survey of the same kids two years later, it was found that all the kids were now in school. Of the kids who were not in school earlier, in one region, 25% were in G now, whereas the rest were enrolled in P; in the second region, all such kids were in G now; while in the third region, 50% of such kids had now joined G while the rest had joined P. As a result, in all three regions put together, 50% of the kids who were earlier out of school had joined G. It was also seen that no surveyed kid had changed schools.

What number of the surveyed kids now were in G in W?

- A 6000
- B 5250
- C 6750
- D 6300

cracku

NE = 7500 W = 12500 S = 10,000

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. They found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

The kids surveyed were further divided into two groups based on whether their mothers had dropped out of school before completing primary education or not. The following table gives the number of kids in different types of schools who dropped out of school before completing primary education:

	G	P	O	Total
NE	500	300	5000	
W	4200	1900	1200	7300
S	5100	300	300	5700
Total	13500	2700	1800	18000

VIDEO SOLUTION

Instructions

Question Numbers: (55 to 58)

In a square layout of site 5m ~ 5m 25 equal-sized square platforms of different heights are built. The heights (in metre) of individual platforms are as shown below:

6	1	2	4	3
9	5	3	2	8
7	8	4	6	5
3	9	5	1	2
1	7	6	3	9

Individuals (all of same height) are seated on these platforms. We say an individual A can reach individual B, if all the three following conditions are met;

- A and B are In the same row or column
- A is at a lower height than B
- If there is/are any individuals (s) between A and B, such individual(s) must be at a height lower than that of A.

Thus in the table given above, consider the Individual seated at height 8 on 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

Question 36

Which of the following is true for any individual at a platform of height 1 m in this layout?

- They can be reached by all the individuals in their own row and column.
- They can be reached by at least 4 individuals.
- They can be reached by at least one individual.
- They cannot be reached by anyone.

cracku

1	6	1	2	4	3
2	9	5	3	2	8
3	7	8	4	6	5
4	3		5	1	2
5	1			3	9

Thus in the table above, consider the individual seated at height 8 in 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

VIDEO SOLUTION

Free Videos for CAT Preparation

Question 37

Which of the following statements is true about this layout?

- A Each row has an individual who can be reached by 5 or more individuals.
- B Each row has an individual who cannot be reached by anyone.
- C Each row has at least two individuals who can be reached by an equal number of individuals.
- D All individuals at the height of 9 m can be reached by at least 5 individuals.

cracku

1	6	1	2	4	3
2	9	5	3	2	8
3	7	8	4	6	5
4	3		5	1	2
5	1			3	9

Thus in the table above, consider the individual seated at height 8 in 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

VIDEO SOLUTION

Instructions

Directions for the following five questions: Answer the following questions based on the information given below:

Abdul, Bikram and Chetan are three professional traders who trade in shares of a company XYZ Ltd. Abdul follows the strategy of buying at the opening of the day at 10 am and selling the whole lot at the close of the day at 3 pm. Bikram follows the strategy of buying at hourly intervals: 10 am, 11am, 12 noon, 1 pm and 2 pm, and selling the whole lot at the close of the day. Further, he buys an equal number of shares in each purchase. Chetan follows a similar pattern as Bikram but his strategy is somewhat different. Chetan's total investment amount is divided equally among his purchases. The profit or loss made by each investor is the difference between the sale value at the close of the day less the investment in purchase. The "return" for each investor is defined as the ratio of the profit or loss to the investment amount expressed as a percentage.

Question 38

One day, two other traders, Dane and Emily joined Abdul, Bikram and Chetan for trading in the shares of XYZ Ltd. Dane followed a strategy of buying equal numbers of shares at 10 am, 11 am and 12 noon, and selling the same numbers at 1 pm, 2 pm and 3 pm. Emily, on the other hand, followed the strategy of buying shares using all her money at 10 am and selling all of them at 12 noon and again buying the shares for all the money at 1 pm and again selling all of them at the close of the day at 3 pm. At the close of the day the following was observed.

- Abdul lost money in the transactions.
- Both Dane and Emily made profits.
- There was an increase in share price during the closing hour compared to the price at 2 pm.
- Share price at 12 noon was lower than the opening price

Which of the following is necessarily false?

- A Share price was at its lowest at 2 pm
- B Share price was at its lowest at 11 am
- C Share price at 1 pm was higher than the share price at 2 pm
- D Share price at 1 pm was higher than the share price at 12 noon
- E more than one of the above

[VIEW SOLUTION](#)

Instructions

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

Question 39

In how many different ways can the ATM serve a customer who gives 500 rupee notes as her preference?

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

In how many different ways can the ATM serve a customer who gives 500 rupee notes as her preference?

	500	200	100
500	510	200	100
	10	2	1
	9	1	3
		0	5



[VIDEO SOLUTION](#)

CAT Percentile Predictor

Instructions

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition				
Ingredient	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

Question 40

The company is planning to launch a balanced diet required for growth needs of adolescent children. This diet must contain at least 30% each of carbohydrate and protein, no more than 25% fat and at least 5% minerals. Which one of the following combinations of equally mixed ingredients is feasible?

- A O and P
- B R and S
- C P and S
- D Q and R
- E O and S

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

For a recuperating patient, the doctor recommended a diet containing at least 30% protein. In how many different ways can we prepare this diet by mixing at least two ingredients?

A) One
B) Three
C) Two
D) None

Handwritten notes on the video solution:
Q & O: 10% protein, 30% protein
S & R: 10% protein, 30% protein
S & P: 10% protein, 30% protein

VIDEO SOLUTION

Instructions

A country has the following types of traffic signals.

3 red lights = stop

2 red lights = turn left

1 red light = turn right

3 green lights = go at 100 km/hr speed

2 green lights = go at 40 km/hr speed

1 green light = go at 20 km/hr speed

A motorist starts at a point on a road and follows all traffic signals. His car is heading towards the north. He encounters the following signals (the time mentioned in each case below is applicable after crossing the previous signal).

Starting point - 1 green light

After half an hour, 1st signal - 2 red and 2 green lights

After 15 min, 2nd signal - 1 red light

After half an hour, 3rd signal - 1 red and 3 green lights

After 24 min, 4th signal - 2 red and 2 green lights

After 15 min, 5th signal - 3 red lights

Question 41

The total distance travelled by the motorist from the starting point till the last signal is

- A 90 km
- B 100 km
- C 120 km
- D None of these

[VIEW SOLUTION](#)

Instructions

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

The kids surveyed were further divided into two groups based on whether their mothers dropped out of school before completing primary education or not.. The table below gives the number of kids in different types of schools for mothers who dropped out of school before completing primary education:

	G	P	O	Total
NE	4200	500	300	5000
W	4200	1900	1200	7300
S	5100	300	300	5700
TOTAL	13500	2700	1800	18000

It is also known that:

1. In S, 60% of the surveyed kids were in G. Moreover, In S, all surveyed kids whose mothers had completed primary education were in school.
2. In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
3. The number of kids in G in NE was the same as the number of kids in G in W.

Question 42

In a follow up survey of the same kids two years later, it was found that all the kids were now in school. Of the kids who were not in school earlier, in one region, 25% were in G now, whereas the rest were enrolled in P; in the second region, all such kids were in G now; while in the third region, 50% of such kids had now joined G while the rest had joined P. As a result, in all three regions put together, 50% of the kids who were earlier out of school

had joined G. It was also seen that no surveyed kid had changed schools.
What percentage of the surveyed kids in S, whose mothers had dropped out before completing primary education, were in G now?

- A 94.7%
- B 89.5%
- C 93.4%
- D Cannot be determined from the given information

cracku

NE = 7500 W = 12500 S = 10,000

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

The kids surveyed were further divided into two groups based on whether their mothers had dropped out of school before completing primary education or not. The following table gives the number of kids in different types of schools who dropped out of school before completing primary education:

	G	P	O	Total
NE	500	300	5000	
W	4200	1900	1200	7300
S	5100	300	300	5700
Total	13500	2700	1800	18000

VIDEO SOLUTION

Important Verbal Ability Questions for CAT (Download PDF)

Instructions

An old woman had the following assets:

- (a) Rs. 70 lakh in bank deposits
- (b) 1 house worth Rs. 50 lakh
- (c) 3 flats, each worth Rs. 30 lakh
- (d) Certain number of gold coins, each worth Rs. 1 lakh

She wanted to distribute her assets among her three children; Neeta, Seeta and Geeta.

The house, any of the flats or any of the coins were not to be split. That is, the house went entirely to one child; a flat went to one child and similarly, a gold coin went to one child.

Question 43

The value of the assets distributed among Neeta, Seeta and Geeta was in the ratio of 1:2:3, while the gold coins were distributed among them in the ratio of 2:3:4. One child got all three flats and she did not get the house.

One child, other than Geeta, got Rs. 30 lakh in bank deposits.

How much did Geeta get in bank deposits (in lakhs of rupees)?

cracku

An old woman had the following assets:

- (a) Rs. 70 lakh in bank deposits
- (b) 1 house worth Rs. 50 lakh
- (c) 3 flats, each worth Rs. 30 lakh
- (d) Certain number of gold coins, each worth Rs. 1 lakh

She wanted to distribute her assets among her three children, Neeta, Seeta and Geeta.

The house, any of the flats or any of the coins were not to be split. That is, the house went entirely to one child, a flat went to one child, and similarly, a gold coin went to one child.

Question (1)

Among the three, Neeta received the least amount in bank deposits, while Geeta received the highest. The value of the assets was distributed equally among the children, as were the gold coins.

How much did Seeta receive in bank deposits (in lakhs of rupees)?

Handwritten notes on the left:

FD - 70
 SD - 50
 3 × 30
 X
 210

Handwritten notes on the right:

N 70
 S 70
 G 70



VIDEO SOLUTION

Instructions

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

A currency exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. The buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The following facts are known about the outlet on a particular day:

1. The amount of L used by the outlet to buy C equals the amount of L it received by selling C.
2. The amounts of L used by the outlet to buy A and B are in the ratio 5:3.
3. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.
4. The outlet received 88000 units of L by selling A during the day.
5. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.
6. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 51000 units of C.

Question 44

What was the buying exchange rate of currency C with respect to currency L on that day?

- A 1.10
- B 0.95
- C 2.20
- D 1.90

cracku

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

An international exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:200:1. The buying exchange rates of each of A, B, and C with respect to L are 5% below their corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The exchange rates are known at the outlet on a particular day:

1. The amount of L received by the outlet to buy C equals the amount of L it received by the outlet to buy A and B are in the ratio 5:3.

2. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

3. The outlet received 8800 units of L by selling A during the day.

4. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.

5. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 4800 units of C.

Handwritten notes on the left:

A → 100k. 95k. 110k.
B → 200k. 114k. 132k.
C → 1k. 0.95k. 1.1k.
Base Buying Selling

	Bought				
A					
B					
C					

VIDEO SOLUTION

Instructions

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

Question 45

What is the maximum number of customers that the ATM can serve with a stock of fifty 500 rupee notes and a sufficient number of notes of other denominations, if all the customers are to be served with at most 20 notes per withdrawal?

- A 12
- B 10
- C 13
- D 16

cracku

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

In how many different ways can the ATM serve a customer who gives 500 rupee notes as her preference?

Handwritten calculations on the right:

500 500 200 100
10 10 2 1
9 1 3
0 5

VIDEO SOLUTION

[Data Interpretation for CAT Questions \(download pdf\)](#)

Instructions

Two students, Amiya and Ramya are the only candidates in an election for the position of class representative. Students will vote based on the intensity level of Amiya's and Ramya's campaigns and the type of campaigns they run. Each campaign is said to have a level of 1 if it is a staid campaign and a level of 2 if it is a vigorous

campaign. Campaigns can be of two types, they can either focus on issues, or on attacking the other candidate.

If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2)\%$, that is, 80% of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and another 10% who would have otherwise voted for Amiya, will not vote at all.
- If Ramya runs a campaign attacking Amiya and Amiya runs a campaign focusing on issues, then 20% of the students who would have otherwise voted for Ramya will vote for Amiya, and another 5% who would have otherwise voted for Ramya, will not vote at all.
- If both run campaigns attacking each other, then 10% of the students who would have otherwise voted for them had they run campaigns focusing on issues, will not vote at all.

Question 46

What is the maximum possible voting margin with which one of the candidates can win?

- A 20%
- B 29%
- C 28%
- D 26%

Two students, Amiya and Ramya are the only candidates in an election for the position of class representative. Students will vote based on the intensity level of Amiya's and Ramya's campaigns and the type of campaigns they run. Each campaign is said to have a level of 1 if it is a staid campaign and a level of 2 if it is a vigorous campaign. Campaigns can be of two types, they can either focus on issues, or on attacking the other candidate.

If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2)\%$, that is, 80% of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and another 10% who would have otherwise voted for Amiya, will not vote at all.
- If Ramya runs a campaign attacking Amiya and Amiya runs a campaign focusing on issues, then 20% of the students who would have otherwise voted for Ramya will vote for Amiya, and another 5% who would have otherwise voted for Ramya, will not vote at all.
- If both run campaigns attacking each other, then 10% of the students who would have otherwise voted for them had they run campaigns focusing on issues, will not vote at all.

A \ R	1	2
1	40% 20% 20%	60% 20% 4%
2	60% 20% 4%	80% 20% 0%

VIDEO SOLUTION

Instructions

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition				
Ingredient	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

Question 47

In what proportion P, Q and S should be mixed to make a diet having at least 60% carbohydrate at the lowest per unit cost?

- A 2 : 1 : 3
- B 4 : 1 : 2
- C 2 : 1 : 4
- D 3 : 1 : 2
- E 4 : 1 : 1

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

For a recuperating patient, the doctor recommended a diet containing at least 30% protein. In how many different ways can we prepare this diet by mixing at least two ingredients?

A) One
C) Three
E) None

Handwritten notes: Q & O mixed 10% 30% Protein. S, R, P < 10%.

VIDEO SOLUTION

Instructions

Question Numbers: (55 to 58)

In a square layout of site 5m ~ 5m 25 equal-sized square platforms of different heights are built. The heights (in metre) of individual platforms are as shown below:

6	1	2	4	3
9	5	3	2	8
7	8	4	6	5
3	9	5	1	2
1	7	6	3	9

Individuals (all of same height) are seated on these platforms. We say an individual A can reach individual B, if all the three following conditions are met;

- (i) A and B are in the same row or column
- (ii) A is at a lower height than B
- (iii) If there is/are any individual(s) between A and B, such individual(s) must be at a height lower than that of A.

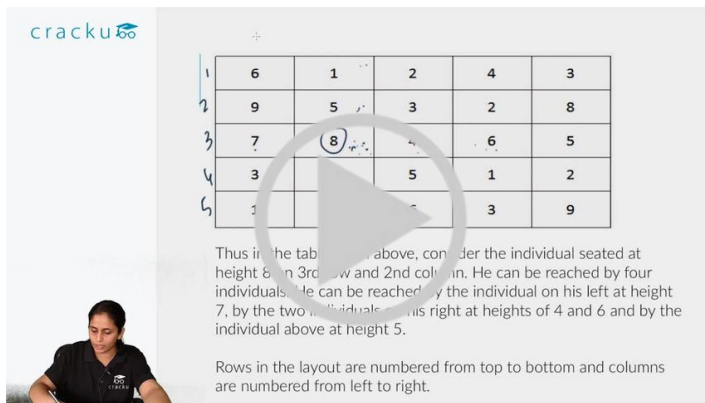
Thus in the table given above, consider the Individual seated at height 8 on 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

Question 48

How many individuals in this layout can be reached by just one individual?

- A 3
- B 5
- C 7
- D 8



Thus in the table given above, consider the individual seated at height 8 on 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

1	6	1	2	4	3
2	9	5	3	2	8
3	7	8	4	6	5
4	3		5	1	2
5	1			3	9

VIDEO SOLUTION

Logical Reasoning for CAT Questions (download pdf)

Instructions

Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

- Column 1 gives the country's identity.
- Column 2 gives the country's GDP as a fraction of the GDP of Country 9.
- Column 3 gives the country's GDP per capita as a fraction of the GDP per capita of Country 10.
- Column 4 gives the country's annual GDP growth rate.
- Column 5 gives the country's annual population growth rate.

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question 49

Which one among the countries 1, 4, 5, and 7 will have the largest population in 2027?

- A Country 1
- B Country 5
- C Country 7
- D Country 4

Out of 10 countries – Country 3 through Country 10 – Country 9 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

- Column 1 gives the country's identity.
- Column 2 gives the country's GDP as a fraction of the GDP of Country 9.
- Column 3 gives the country's GDP per capita as a fraction of the per capita of Country 10.
- Column 4 gives the country's annual GDP growth rate.
- Column 5 gives the country's annual population growth rate.

Country	GDP	GDP per capita	GDP growth rate	Pop. growth rate
Country 1	0.15	0.25	2.5%	1.5%
Country 2	0.20	0.30	3.0%	2.0%
Country 3	0.25	0.35	3.5%	2.5%
Country 4	0.30	0.40	4.0%	3.0%
Country 5	0.35	0.45	4.5%	3.5%
Country 6	0.40	0.50	5.0%	4.0%
Country 7	0.45	0.55	5.5%	4.5%
Country 8	0.50	0.60	6.0%	5.0%

Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question (1)

Which one among the countries 1 through 8, has the smallest population in 2024?

A) Country 8
B) Country 3
C) Country 7
D) Country 5

Population ~ GDP

VIDEO SOLUTION

Instructions

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition				
Ingredient	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

Question 50

Which among the following is the formulation having the lowest cost per unit for a diet having 10% fat and at least 30% protein? The diet has to be formed by mixing two ingredients.

- A P and Q
- B P and S
- C P and R
- D Q and S
- E R and S

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (Q, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is Q: 150, P: 50, Q: 200, R: 500, S: 100.

Ingredient	Carbohydrate %	Protein %	Fat %	Minerals %
Q	45	50	0	5
P	5	50	40	0
Q	10	50	50	0
R	80	20	0	0
S	50	50	20	20

For a recuperating patient, the doctor recommended a diet containing at least 30% protein. In how many different ways can we prepare this diet by mixing at least two ingredients?

A) One
B) Two
C) Three
D) Four
E) None

Handwritten notes: Q & P, 10%, 30%, 10%, 0%, 10%, 50%, 10%

VIDEO SOLUTION

Instructions

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

A currency exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. The buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The following facts are known about the outlet on a particular day:

1. The amount of L used by the outlet to buy C equals the amount of L it received by selling C.
2. The amounts of L used by the outlet to buy A and B are in the ratio 5:3.
3. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.
4. The outlet received 88000 units of L by selling A during the day.
5. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.
6. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 51000 units of C.

Question 51

How many units of currency A did the outlet buy on that day?

Handwritten notes:
A → 100k, 95k, 110k, 132k
B → 120k, 114k, 132k, 144k
C → 1k, 0.95k, 1.1k, 1.2k
Base Buying Selling

	Bought				
A					
B					
C					

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

A currency exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. The buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The following facts are known about the outlet on a particular day:

1. The amount of L used by the outlet to buy C equals the amount of L it received by selling C.
2. The amounts of L used by the outlet to buy A and B are in the ratio 5:3.
3. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.
4. The outlet received 88000 units of L by selling A during the day.
5. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.
6. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 51000 units of C.

VIDEO SOLUTION

Quantitative Aptitude for CAT Questions (download pdf)

Instructions

There are only four brands of entry level smartphones called Azra, Bysi, Cxqi, and Dipq in a country. Details about their market share, unit selling price, and profitability (defined as the profit as a percentage of the revenue) for the year 2016 are given in the table below:

Brand	Market Share (%)	Unit Selling Price (Rs.)	Profitability (%)
Azra	40	15,000	10
Bysi	25	20,000	30
Cxqi	15	30,000	40
Dipq	20	25,000	30

In 2017, sales volume of entry level smartphones grew by 40% as compared to that in 2016. Cxqi offered a 40% discount on its unit selling price in 2017, which resulted in a 15% increase in its market share. Each of the other three brands lost 5% market share. However, the profitability of Cxqi came down to half of its value in 2016. The unit selling prices of the other three brands and their profitability values remained the same in 2017 as they were in 2016.

Question 52

The brand that had the highest profit in 2016 is:

- A Bysi
- B Dipq
- C Cxqi
- D Azra



VIDEO SOLUTION

Question 53

The complete list of brands whose profits went up in 2017 from 2016 is:

- A Azra, Bysi, Dipq
- B Cxqi, Azra, Dipq
- C Azra, Bysi, Cxqi
- D Bysi, Cxqi, Dipq

Details about the market share, unit selling price, and profitability (calculated as the profit as a percentage of the revenue) for the year 2016 are given in the table below:

Brand	Market Share (%)	Unit Selling Price (Rs.)	Profitability (%)
Axx	25	15,000	30
Bxx	35	10,000	20
Cxx	40	8,000	15
Dxx	20	20,000	40

In 2017, sales volume of entry level smartphones grew by 40% as compared to that in 2016. Cxx offered a 40% discount on its unit selling price in 2017, which resulted in a 15% increase in its market share. Each of the other three brands lost 5% market share. However, the profitability of Cxx came down to half of its value in 2016. The unit selling prices of the other three brands and their profitability values remained the same in 2017 as they were in 2016.

2017 → 140

A
B
C
D

VIDEO SOLUTION

Instructions

An old woman had the following assets:

- (a) Rs. 70 lakh in bank deposits
- (b) 1 house worth Rs. 50 lakh
- (c) 3 flats, each worth Rs. 30 lakh
- (d) Certain number of gold coins, each worth Rs. 1 lakh

She wanted to distribute her assets among her three children; Neeta, Seeta and Geeta.

The house, any of the flats or any of the coins were not to be split. That is, the house went entirely to one child; a flat went to one child and similarly, a gold coin went to one child.

Question 54

The value of the assets distributed among Neeta, Seeta and Geeta was in the ratio of 1:2:3, while the gold coins were distributed among them in the ratio of 2:3:4. One child got all three flats and she did not get the house.

One child, other than Geeta, got Rs. 30 lakh in bank deposits.

How many gold coins did the old woman have?

- A 72
- B 90
- C 180
- D 216

cracku

An old woman had the following assets:

- (a) Rs. 70 lakh in bank deposits
- (b) 1 house worth Rs. 50 lakh
- (c) 3 flats, each worth Rs. 30 lakh
- (d) Certain number of gold coins, each worth Rs. 1 lakh

She wanted to distribute her assets among her three children; Neeta, Seeta and Geeta.

The house, any of the flats or any of the coins were not to be split. That is, the house went entirely to one child; a flat went to one child and similarly, a gold coin went to one child.

Question (1)

Among the three, Neeta received the least amount in bank deposits, while Geeta received the highest. The value of the assets was distributed equally among the children, as were the gold coins.

How much did Seeta receive in bank deposits (in lakhs of rupees)?

N 10 S 70 G 10

70 - BD
50 - House
3 × 30 - Flats
X - Gold Coins
210

VIDEO SOLUTION

Know the CAT Percentile Required for IIM Calls

Answers

1.A	2.D	3.B	4.A	5.C	6.A	7.2	8.C
9.C	10.E	11.A	12.A	13.C	14.A	15.8	16.1
17.9	18.7	19.6	20.240	21.A	22.A	23.2	24.C
25.13	26.A	27.0	28.B	29.A	30.6	31.A	32.3
33.D	34.E	35.A	36.D	37.C	38.E	39.7	40.E
41.A	42.A	43.20	44.D	45.A	46.B	47.E	48.C
49.A	50.D	51.1200	52.C	53.A	54.B		

Explanations

1. A

Amiya runs a campaign on issues, and we need to find the maximum vote share that she can get.

We are trying to maximise the number of voters for Amiya, that means Amiya needs to run a vigorous campaign.

Since, we are trying to increase the vote share for Amiya, we want as many voters as possible transferred from Ramya's share to Amiya's votes.

As the number of votes a candidate receives is proportional to the intensity level of the campaign, we want Ramya also to run a vigorous AND attacking campaign, so that her votes are transferred to Amiya.

In this scenario we have $20 \times (2+2)\%$ of the people voting, 80% of the people. Of that, if both had ran issues campaign, they would have each received 40% of the votes.

Now, we want Ramya to run an attacking campaign, where 20% of the people that would have voted for her vote for Amiya.

So 20% of 40% of the votes are transferred to Amiya. That is 8% of the votes. And we are also told that, 5% that would have voted for her dont vote, so 5% of 40% dont vote, that is 2% of the voters.

Final Tally is Amiya gets 48% of the votes, and Ramya gets 30% of the votes.

 VIDEO SOLUTION

2. D

If both Ramya and Amiya run staid campaigns, the intensity of each staid campaign is 1

So total number of voters that would vote for them if they focused on issues will be $20 \times (1+1)\% = 40\%$

This means if they had both ran regarding issues, then they would get 20% of votes each.

We are told that both of them run attacking campaigns,

And the rule for mutual attacking campaign is 10% of voters who would have voted for each candidate will not vote.

That means 10% of 20% of each candidate will not vote, now that it is a mutually attacking campaign.

That means, each candidate receives 18% of the votes.

Total votes received is 36%.

 VIDEO SOLUTION

3. B

We are looking for the minimum possible number of votes that Ramya can get when she runs an attacking campaign.

To minimise the number of votes, we can have Ramya run a staid campaign to minimise the votes, so minimum intensity, which will get her 20% of the votes if she ran with issues. Now that she is running with attacking, she will loose 20% of the votes to Amiya and 5% of the votes will not vote anymore.

That is a total 25% loss. Remaining votes she will get is 75% of the 20% which will leave her with 15% of the votes.

 VIDEO SOLUTION

4. **A**

Let the share prices at 10,11,12,1,2,3 be p,q,r,s,t,u

Since Abdul lost all money, $p > u$

From Dane it can be observed, $s+t+u > p+q+r$

$p > r$ and $u > t$

From Emily it can be observed, $r+u > p+s$

From these inequalities it can be inferred that the p is the highest.

 VIEW SOLUTION

5. **C**

Check out the options.

In last row, only one individual can't be reached by anyone.

In the fourth row, only one individual can't be reached by anyone.

In the middle column, every individual can be reached by atleast another individual.

In the fourth column, both 1 and 2 cannot be reached by any person. Hence, fourth column is the correct answer.

 VIDEO SOLUTION

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6. **A**

Let us make note of the information given in the set.

The set states that rural kids were surveyed from 3 regions NE, West and South.

50 students each were surveyed from 150 villages in the NE, 250 villages from the West and 200 villages from the South.

Total number of kids from the NE = $150 \times 50 = 7500$

Total number of kids from the West = $250 \times 50 = 12500$

Total number of kids from the South = $200 \times 50 = 10,000$.

The table, given in the question, gives the number of students whose mothers dropped out before completing primary education. Therefore, the mothers of the remaining students should have completed primary education.

There are 7500 students from the NE in total. The mothers of 5000 of those students dropped out before completing primary education. Therefore, the remaining 2500 kids should have mothers who have completed primary education.

Let us make 2 tables - one representing the number of students and the other representing the number of students whose mother has completed primary education.

55% of the surveyed kids studied in schools run by the Government.

Number of kids studying in Govt. School should be $0.55 \times 30000 = 16,500$

From the given table, we know that the mothers of 13,500 kids who went to Govt. Schools dropped out of primary school.

Therefore, the remaining $16,500 - 13,500 = 3000$ kids should have mother who have completed primary school.

37% of the surveyed kids study in private schools.

Number of kids studying in private schools should be $0.37 \times 30000 = 11,100$.

Number of kids in private schools whose mothers have completed primary school = $11,100 - 2700 = 8400$.

8% of the surveyed kids did not go to school.

Number of kids who did not go to school = $0.08 \times 30000 = 2400$

Number of kids not going to schools whose mothers have completed primary school = $2400 - 1800 = 600$.

In S, 60% of the surveyed kids were in G.

Therefore, 6000 kids in S must be from G.

In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.

There are 300 O kids whose mothers dropped out before completing primary education. These kids represent 50% of the total number of O-kids. Therefore, there must be 600 O-kids from NE - 300 kids should have mothers who dropped out before completing primary education and 300 kids should have mothers who have completed primary education.

The number of kids in G in NE was the same as the number of kids in G in W. Therefore, the number of kids in G in NE and the number of kids in G in W should be equal to 5250. Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250			12500
S	6000			10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050			5200
S	900			4300
Total	3000	8400	600	12000

We have been given that in S, all surveyed kids whose mothers had completed primary education were in school. Therefore, the number of kids not going to school whose mothers have completed primary education in S should be 0. Filling the tables, we get,

Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250	5750	1500	12500
S	6000	3700	300	10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050	3850	300	5200
S	900	3400	0	4300
Total	3000	8400	600	12000

There are 300 students who are not in school in W whose mothers have completed primary education.

Therefore, option A is the right answer.

▶ VIDEO SOLUTION

7.2

We can make the following table from "the total amount of money kept in the three pouches in the first column of the third row is Rs. 4."

If the minimum and maximum value are 1, then the sum of the three pouches in the middle will be Rs 3.

	Column 1	Column 2	Column 3
Row 1			
Row 2		3 (1,1,1)	
Row 3	4 (1,1,2)		

If we calculate the maximum and minimum value possible for each slot in column 1. For the slot, column 1 and row 1, the maximum value possible is 10{2,4,4} while the minimum value possible is 8{2,2,4}.

Similarly, for the slot, column 1 and row 2, the maximum value possible is 13{3,5,5} while the minimum value possible is 11{3,3,5}.

It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. Thus the sum of coins in a row or column must be a multiple of 9.

So, we can iterate that 10,13,4 ...{27} is the only sum possible for the slots of column 1.

We now know two elements of row 2, thus we can iterate from the maximum and the minimum value possible for the slot {column 3, row 2} that 38 is the only value possible for the slot.

We can make the following table:

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)		
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)		

Similarly, we can find the amount for Column 2.

For the slot, column 2 and row 1, the maximum value possible is 22{6,8,8} while the minimum value possible is 20{6,6,8}.

For the slot, column 2 and row 3, the maximum value possible is 5{1,2,3} while the minimum value possible is 4{1,1,2}.

Thus {20,3,4} is the only solution possible.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	

We can similarly make the following table for the last column.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	6 (1,2,3)
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	10 (2,3,5)

Answer 2

[VIDEO SOLUTION](#)

8. C

Let '100x' be the number of smartphones sold in year 2016.

Total revenue generated by Azra = $40x \times 15000$ = Rs. 600000x

Total revenue generated by Bysi = $25x \times 20000$ = Rs. 500000x

Total revenue generated by Cxqi = $15x \times 30000$ = Rs. 450000x

Total revenue generated by Dipq = $20x \times 25000$ = Rs. 500000x

We can see that revenue generated by Azra is the highest among all four brands. Hence, option C is the correct answer.

[VIDEO SOLUTION](#)

9. C

If at the starting point, the car was heading towards south then following would have occurred : 1st 10 km in south ,2nd 10 km east, 3rd 20 km in south ,4th 40 km west, 5th 10 km in south so overall 30 km to the west and 40 km to the south from starting point.

[VIEW SOLUTION](#)

10. E

Similar is the case with this question. Nothing can be inferred as we do not know about the fluctuation of share prices.

[VIEW SOLUTION](#)

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11. A

Abdul gets the highest return since he buys all the shares at the start of the day.

Let Bikram and Chetan spend equal amounts of money. Since Chetan spends his money equally on all business, he would buy more of a shares at lower price and less shares at a higher price. On the other hand, Bikram buys an equal number of shares at each price. So, the return obtained by Chetan would be more than the return obtained by Bikram. Option a) is the correct answer.

[VIEW SOLUTION](#)

12. A

Now if after starting point at first point the 1st signal were 1 red and 2 green lights then distance traveled by the car would be as follows : 10 km in north then 10 km east then 20 km south then 40 km to west then 10 km to south . So overall the position of car is 30 km to the west and 20 km to south as compared to original position.

[VIEW SOLUTION](#)

13. C

According to given conditions, the car first travels 10 km north then 10 km to west then gain 20 km north then further travels east for 40 km then travel north for another 10 km before finally stopping. So in all the car travels 30 km to the east and 40 km north. Hence, calculating the distance radially we get 50 km in North-East direction.

[VIEW SOLUTION](#)

14. A

We want a diet with 10% minerals and at least 30% protien. The mixture of O and Q is suitable because both contain 10% minerals. In required proportion we can get 10% minerals and at least 30% proteins.

Solution P and R contains minerals which are less than 10% and maximum mineral %age in any component is 10. So if we include P or R in any mixture, we can't get 10% mineral %age.

So only 1 solution of O and Q is possible.

[VIDEO SOLUTION](#)

15. 8

We can make the following table from "the total amount of money kept in the three pouches in the first column of the third row is Rs. 4."

If the minimum and maximum value are 1, then the sum of the three pouches in the middle will be Rs 3.

	Column 1	Column 2	Column 3
Row 1			
Row 2		3 (1,1,1)	
Row 3	4 (1,1,2)		

If we calculate the maximum and minimum value possible for each slot in column 1. For the slot, column 1 and row 1, the maximum value possible is 10{2,4,4} while the minimum value possible is 8{2,2,4}.

Similarly, for the slot, column 1 and row 2, the maximum value possible is 13{3,5,5} while the minimum value possible is 11{3,3,5}.

It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. Thus the sum of coins in a row or column must be a multiple of 9.

So, we can iterate that 10,13,4 ...{27} is the only sum possible for the slots of column 1.

We now know two elements of row 2, thus we can iterate from the maximum and the minimum value possible for the slot {column 3, row 2} that 38 is the only value possible for the slot.

We can make the following table:

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)		
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)		

Similarly, we can find the amount for Column 2.

For the slot, column 2 and row 1, the maximum value possible is 22{6,8,8} while the minimum value possible is 20{6,6,8}.

For the slot, column 2 and row 3, the maximum value possible is 5{1,2,3} while the minimum value possible is 4{1,1,2}.

Thus {20,3,4} is the only solution possible.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	

We can similarly make the following table for the last column.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	6 (1,2,3)
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	10 (2,3,5)

Answer 8

[VIDEO SOLUTION](#)

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16.1

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

The value of F can only be 0 as $F+F=F$ can only hold if $F=0$.

Also, A can only be 1(in the second column) because to get a **carry** of more than 1, B has to be a double-digit number which is not possible. (**A carry is a digit that is transferred from one column of digits to another column of more significant digits.**)

So the data can be tabulated as follows:

		B	H	1	1	G	0
+		1	H	J	0	K	0
	1	1	0	G	C	1	0

Since the last row in the third column is 0, the carry to the second column must have been 1, Hence $B+1+1=11 \Rightarrow B=9$

In the 4th column, $H+H = 10$ since a carry 1 has gone to the 3rd column. Hence $H=5$.

$G+K$ must be 11 and the carry 1 goes to the next column, so $C=1+1=2$.

Now, G,K can take values (3,8), (4,7) and (5,6) in any order.

From 5th column $G=J+1 \Rightarrow J=G-1$

Case: $G=3$ and $K=8$, here $J=2$ which is not possible as $C=2$

Case: $G=8$ and $K=3$, $J=7$, a possible case.

Case: $G=4$ and $K=7$, $J=3$ possible

Case: $G=7$ and $K=4$, $J=6$ possible

Case: $G=5$ and $K=6$, $J=4$ not possible as $H=5$.

Case: $G=6$ and $K=5$, $J=5$ both J and K are same, not possible.

Hence the cases can be tabulated as follows:

		9	5	1	1	8	0
+		1	5	7	0	3	0
	1	1	0	8	2	1	0

		9	5	1	1	7	0
+		1	5	6	0	4	0
	1	1	0	7	2	1	0

		9	5	1	1	4	0
+		1	5	3	0	7	0
	1	1	0	4	2	1	0

The letter A represents 1.

 VIDEO SOLUTION

17.9

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

The value of F can only be 0 as $F+F=F$ can only hold if $F=0$.

Also, A can only be 1(in the second column) because to get a **carry** of more than 1, B has to be a double-digit number which is not possible. (**A carry is a digit that is transferred from one column of digits to another column of more significant digits.**)

So the data can be tabulated as follows:

		B	H	1	1	G	0
+		1	H	J	0	K	0
	1	1	0	G	C	1	0

Since the last row in the third column is 0, the carry to the second column must have been 1, Hence $B+1+1=11$
 $\Rightarrow B=9$

In the 4th column, $H+H = 10$ since a carry 1 has gone to the 3rd column. Hence $H=5$.

$G+K$ must be 11 and the carry 1 goes to the next column, so $C=1+1=2$.

Now, G, K can take values (3,8), (4,7) and (5,6) in any order.

From 5th column $G=J+1 \Rightarrow J=G-1$

Case: $G=3$ and $K=8$, here $J=2$ which is not possible as $C=2$

Case: $G=8$ and $K=3$, $J=7$, a possible case.

Case: $G=4$ and $K=7$, $J=3$ possible

Case: $G=7$ and $K=4$, $J=6$ possible

Case: $G=5$ and $K=6$, $J=4$ not possible as $H=5$.

Case: $G=6$ and $K=5$, $J=5$ both J and K are same, not possible.

Hence the cases can be tabulated as follows:

		9	5	1	1	8	0
+		1	5	7	0	3	0
	1	1	0	8	2	1	0

		9	5	1	1	7	0
+		1	5	6	0	4	0
	1	1	0	7	2	1	0

		9	5	1	1	4	0
+		1	5	3	0	7	0
	1	1	0	4	2	1	0

The letter B represents 9.

[VIDEO SOLUTION](#)

18.7

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

The value of F can only be 0 as $F+F=F$ can only hold if $F=0$.

Also, A can only be 1 (in the second column) because to get a **carry** of more than 1, B has to be a double-digit number which is not possible. (**A carry is a digit that is transferred from one column of digits to another column of more significant digits.**)

So the data can be tabulated as follows:

		B	H	1	1	G	0
+		1	H	J	0	K	0
	1	1	0	G	C	1	0

Since the last row in the third column is 0, the carry to the second column must have been 1, Hence $B+1+1=11$
 $\Rightarrow B=9$

In the 4th column, $H+H = 10$ since a carry 1 has gone to the 3rd column. Hence $H=5$.

$G+K$ must be 11 and the carry 1 goes to the next column, so $C=1+1=2$.

Now, G, K can take values (3,8), (4,7) and (5,6) in any order.

From 5th column $G=J+1 \Rightarrow J=G-1$

Case: $G=3$ and $K=8$, here $J=2$ which is not possible as $C=2$

Case: $G=8$ and $K=3$, $J=7$, a possible case.

Case: $G=4$ and $K=7$, $J=3$ possible

Case: $G=7$ and $K=4$, $J=6$ possible

Case: $G=5$ and $K=6$, $J=4$ not possible as $H=5$.

Case: $G=6$ and $K=5$, $J=5$ both J and K are same, not possible.

Hence the cases can be tabulated as follows:

		9	5	1	1	8	0
+		1	5	7	0	3	0
	1	1	0	8	2	1	0

		9	5	1	1	7	0
+		1	5	6	0	4	0
	1	1	0	7	2	1	0

		9	5	1	1	4	0
+		1	5	3	0	7	0
	1	1	0	4	2	1	0

In all possible cases 7 is already represented by a letter other than D. Hence 7 is the answer.

 VIDEO SOLUTION

19.6

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

The value of F can only be 0 as $F+F=F$ can only hold if $F=0$.

Also, A can only be 1 (in the second column) because to get a **carry** of more than 1, B has to be a double-digit number which is not possible. (**A carry is a digit that is transferred from one column of digits to another column of more significant digits.**)

So the data can be tabulated as follows:

		B	H	1	1	G	0
+		1	H	J	0	K	0
	1	1	0	G	C	1	0

Since the last row in the third column is 0, the carry to the second column must have been 1, Hence $B+1+1=11 \Rightarrow B=9$

In the 4th column, $H+H=10$ since a carry 1 has gone to the 3rd column. Hence $H=5$.

$G+K$ must be 11 and the carry 1 goes to the next column, so $C=1+1=2$.

Now, G, K can take values (3,8), (4,7) and (5,6) in any order.

From 5th column $G=J+1 \Rightarrow J=G-1$

Case: $G=3$ and $K=8$, here $J=2$ which is not possible as $C=2$

Case: $G=8$ and $K=3$, $J=7$, a possible case.

Case: $G=4$ and $K=7$, $J=3$ possible

Case: $G=7$ and $K=4$, $J=6$ possible

Case: $G=5$ and $K=6$, $J=4$ not possible as $H=5$.

Case: $G=6$ and $K=5$, $J=5$ both J and K are same, not possible.

Hence the cases can be tabulated as follows:

		9	5	1	1	8	0
+		1	5	7	0	3	0
	1	1	0	8	2	1	0

		9	5	1	1	7	0
+		1	5	6	0	4	0
	1	1	0	7	2	1	0

		9	5	1	1	4	0
+		1	5	3	0	7	0
	1	1	0	4	2	1	0

From the table it is clear that 6 cannot be represented by G.

VIDEO SOLUTION

20.240

It is given that the base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. Let us assume that base exchange rates are '100a', '120a' and 'a' in that order.

It is given that the buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates. Therefore, we can say that the buying exchange rates are 95a, 114a, 0.95a.

It is given that the selling exchange rates of each of A, B, and C with respect to L are 10% above their corresponding base exchange rates. Therefore, we can say that the selling exchange rates are 110a, 132a, 1.1a.

We know about the opening and closing units in stock for each currency. Let us draw the table accordingly.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	95a	100a	110a	2500			3300
B	114a	120a	132a	4800			4800
C	0.95a	a	1.1a	48000			51000

Let 'p', 'q' and 'r' be the number of units of currency A, B and C bought by the outlet on that day.

Then, we can say that the outlet sold 'p - 800', 'q' and 'r - 3000' units of currency A, B and C respectively.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	95a	100a	110a	2500	p	p - 800	3300
B	114a	120a	132a	4800	q	q	4800
C	0.95a	a	1.1a	48000	r	r - 3000	51000

It is given that the amount of L used by the outlet to buy C equals the amount of L it received by selling C.

$$\Rightarrow 0.95a \cdot r = 1.1a \cdot (r - 3000)$$

$$\Rightarrow 0.15r = 3300$$

$$\Rightarrow r = 22000$$

It is also given that the amounts of L used by the outlet to buy A and B are in the ratio 5:3.

$$\Rightarrow \frac{p \cdot 95a}{q \cdot 114a} = \frac{5}{3}$$

$$\Rightarrow p = 2q$$

Also, the amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

$$\Rightarrow \frac{(p - 800) \cdot 110a}{q \cdot 132a} = \frac{5}{9}$$

$$\Rightarrow \frac{(2q - 800) \cdot 110a}{q \cdot 132a} = \frac{5}{9}$$

$$\Rightarrow q = 600$$

$$\text{Therefore, } p = 2q = 2 \times 600 = 1200.$$

It is given that the outlet received 88000 units of L by selling A during the day.

$$\Rightarrow (p-800) \times 110a = 88000$$

$$\Rightarrow (1200-800) \times 110a = 88000$$

$$\Rightarrow 44000a = 88000$$

$$\Rightarrow a = 2$$

We can fill the entire table and answer all the questions.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	190	200	220	2500	1200	400	3300
B	228	240	264	4800	600	600	4800
C	1.9	2	2.2	48000	22000	19000	51000

From the table we can see that the base exchange rate of currency B with respect to currency L was 240.

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21. A

Let '100x' be the number of smartphones sold in year 2016. Then the number of smartphones sold in 2017 = $1.4 \times 100x = 140x$

It is given that Cxqi offered a 40% discount on its unit selling price in 2017 i.e. selling price in 2017 = $0.6 \times 30000 = \text{Rs. } 18000$

Also Cxqi's market share increased by 15% whereas the other three brands lost 5% market share.

Brand	Market Share (%)	No of units	Unit Selling Price (Rs.)	Profitability(%)
Azra	35	49x	15,000	10
Bysi	20	28x	20,000	30
Cxqi	30	42x	18,000	20
Dipq	15	21x	25,000	30

$$\text{Amount of profit generated by Azra} = \frac{10}{100} \times 15000 \times 49x = 73500x$$

$$\text{Amount of profit generated by Bysi} = \frac{30}{100} \times 20000 \times 28x = 168000x$$

$$\text{Amount of profit generated by Cxqi} = \frac{20}{100} \times 18000 \times 42x = 151200x$$

$$\text{Amount of profit generated by Dipq} = \frac{30}{100} \times 25000 \times 21x = 157500x$$

We can see that brand Bysi generated maximum profit in year 2017. Hence, option A is the correct answer.

[VIDEO SOLUTION](#)

22. A

We are given the following table,

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

This table compares the GDP, GDP per capita, and population of eight countries, all of which are referenced using a consistent set of units.

Specifically, the GDP values are all given with the same country as the reference point (The reference point being Country 9)

Similarly, the GDP per capita among the eight countries in the table is given using the same reference point (The reference point being Country 10)

As a result, when comparing two countries from the list in terms of GDP, GDP per capita, or population, we can directly use the values presented in the table, since they are all based on a uniform reference point.

We can compare the population of two countries using the formula, $Population = \frac{GDP}{GDP\ Per\ Capita}$

For example, to compare the GDP of Country A and Country B, we can simply use the values listed under "GDP" in the table, as both are referenced using the same country throughout. This eliminates the need for additional conversions or adjustments.

We are given the GDP growth rate and we are told it is constant over the next three years,

Country 4 GDP: 0.12 and the given growth rate is 0.5%

GDP of Country 4 in 2026 will be $0.12 \times (1.005)^2$ this equals 0.121203

Country 5 GDP: 0.1 and the given growth rate is 3.2

GDP of Country 5 in 2026 will be $0.1 \times (1.007)^2$ this equals 0.1014049

Ratio will be, $\frac{0.121203}{0.1065024}$ this equals 1.19523

Hence, the answer is Option A

[VIDEO SOLUTION](#)

23.2

Neeta received least amount in bank deposits implies she received highest amount in property and the vice-versa for Geeta. The assets are 3 flats worth 90 lakh, a house worth 50 lakh, and a deposit worth 70 lakh. The total value of assets is 210 lakhs. They are divided equally, so each will receive assets worth 70 lakh.

No one daughter can get 3 flats as the total value of asset will be 90 lakhs which is greater than actual share.

All three daughters can't get 1-1 flat each as well. In that case, the daughter who owns 1 flat and the house will have assets worth $30+50 = 80$ lakhs which is more than the actual share. Hence, we can conclude the one of the three daughter gets 2 flats and 10 lakhs bank deposit.

Out of the remaining two daughters, one will get the house and bank deposit worth 20 lakhs and the other one must have 1 flat and 40 lakhs in bank deposit. On the basis of bank distribution we can easily determine that property and bank deposits for each Neera, Seeta and Geeta.

	Neeta	Seeta	Geeta
Property	2 flats worth 60 lakhs	house worth 50 lakhs	1 flat worth 30 lakhs
Bank deposits	10 lakhs	20 lakhs	40 lakhs

From the table, we can see that Neeta received 2 flats.

[VIDEO SOLUTION](#)

24. C

Neeta received least amount in bank deposits implies she received highest amount in property and the vice-versa for Geeta. The assets are 3 flats worth 90 lakh, a house worth 50 lakh, and a deposit worth 70 lakh. The total value of assets is 210 lakhs. They are divided equally, so each will receive assets worth 70 lakh.

No one daughter can get 3 flats as the total value of asset will be 90 lakhs which is greater than actual share.

All three daughters can't get 1-1 flat each as well. In that case, the daughter who owns 1 flat and the house will have assets worth $30+50 = 80$ lakhs which is more than the actual share. Hence, we can conclude the one of the three daughter gets 2 flats and 10 lakhs bank deposit.

Out of the remaining two daughters, one will get the house and bank deposit worth 20 lakhs and the other one must have 1 flat and 40 lakhs in bank deposit. On the basis of bank distribution we can easily determine that property and bank deposits for each Neeta, Seeta and Geeta.

	Neeta	Seeta	Geeta
Property	2 flats worth 60 lakhs	house worth 50 lakhs	1 flat worth 30 lakhs
Bank deposits	10 lakhs	20 lakhs	40 lakhs

From the table, we can see that Seeta must have received Rs. 20 lakh in bank deposits. Hence, option C is the correct answer.

 VIDEO SOLUTION

25. 13

We can make the following table from "the total amount of money kept in the three pouches in the first column of the third row is Rs. 4."

If the minimum and maximum value are 1, then the sum of the three pouches in the middle will be Rs 3.

	Column 1	Column 2	Column 3
Row 1			
Row 2		3 (1,1,1)	
Row 3	4 (1,1,2)		

If we calculate the maximum and minimum value possible for each slot in column 1. For the slot, column 1 and row 1, the maximum value possible is 10{2,4,4} while the minimum value possible is 8{2,2,4}.

Similarly, for the slot, column 1 and row 2, the maximum value possible is 13{3,5,5} while the minimum value possible is 11{3,3,5}.

It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. Thus the sum of coins in a row or column must be a multiple of 9.

So, we can iterate that 10,13,4 ...{27} is the only sum possible for the slots of column 1.

We now know two elements of row 2, thus we can iterate from the maximum and the minimum value possible for the slot {column 3, row 2} that 38 is the only value possible for the slot.

We can make the following table:

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)		
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)		

Similarly, we can find the amount for Column 2.

For the slot, column 2 and row 1, the maximum value possible is 22{6,8,8} while the minimum value possible is 20{6,6,8}.

For the slot, column 2 and row 3, the maximum value possible is 5{1,2,3} while the minimum value possible is 4{1,1,2}.

Thus {20,3,4} is the only solution possible.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	

We can similarly make the following table for the last column.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	6 (1,2,3)
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	10 (2,3,5)

The total amount of money (in rupees) in the three pouches kept in the first column of the second row=13

Correct answer 13

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26. A

We are given the following table,

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

This table compares the GDP, GDP per capita, and population of eight countries, all of which are referenced using a consistent set of units.

Specifically, the GDP values are all given with the same country as the reference point (The reference point being Country 9)

Similarly, the GDP per capita among the eight countries in the table is given using the same reference point (The reference point being Country 10)

As a result, when comparing two countries from the list in terms of GDP, GDP per capita, or population, we can directly use the values presented in the table, since they are all based on a uniform reference point.

We can compare the population of two countries using the formula, $Population = \frac{GDP}{GDP\ Per\ Capita}$

For example, to compare the GDP of Country A and Country B, we can simply use the values listed under "GDP" in the table, as both are referenced using the same country throughout. This eliminates the need for additional conversions or adjustments.

In this question we are asked to find which of the countries among the four options has the least population in 2024 (The current year)

Option A: Country 8

GDP=0.07 and GDP Per Capita=0.41

Population=0.07/0.41 or $\frac{7}{41}$

Option B: Country 3

GDP=0.13 and GDP Per Capita=0.02

Population=0.13/0.02 or $\frac{13}{2}$

Option C: Country 7

GDP=0.08 and GDP Per Capita=0.3

Population=0.08/0.3 or $\frac{8}{30}$

Option D: Country 5

GDP=0.1 and GDP Per Capita=0.36

Population=0.1/0.36 or $\frac{10}{36}$

Comparing the four values, we are looking for the smallest values among the four,

$\frac{7}{41}$, $\frac{13}{2}$, $\frac{8}{30}$, $\frac{10}{36}$

$\frac{7}{41}$ is the smallest value among the four, hence Option A or Country 8 is the answer.

 VIDEO SOLUTION

27.0

We are given the following table,

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

This table compares the GDP, GDP per capita, and population of eight countries, all of which are referenced using a consistent set of units.

Specifically, the GDP values are all given with the same country as the reference point (The reference point being Country 9)

Similarly, the GDP per capita among the eight countries in the table is given using the same reference point (The reference point being Country 10)

As a result, when comparing two countries from the list in terms of GDP, GDP per capita, or population, we can directly use the values presented in the table, since they are all based on a uniform reference point.

We can compare the population of two countries using the formula, $Population = \frac{GDP}{GDP\ Per\ Capita}$

For example, to compare the GDP of Country A and Country B, we can simply use the values listed under "GDP" in the table, as both are referenced using the same country throughout. This eliminates the need for additional conversions or adjustments.

We are asked to find the number of countries where the GDP per capita is lower in 2027 than it was in 2024.

For the GDP per capita to be lower in the consequent years, the population growth rate has to exceed the GDP growth rate, since GDP per capita is nothing but GDP divided by the population. That means if the population growth rate is lesser than that of the GDP growth rate, the GDP per capita will only increase.

We can clearly see that none of the following countries fall in the scenario where GDP growth rate is lesser than that of the population growth rate. Countries 1, 2 and 7 have actually decreasing population rates thereby definitely increasing the GDP per capita.

The rest of the countries have GDP growth rates larger than the population growth rates.

Hence, we can conclude that, none of the countries will have a smaller GDP per capita in 2027 when compared to 2024.

▶ VIDEO SOLUTION

28. B

It is given that the base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. Let us assume that base exchange rates are '100a', '120a' and 'a' in that order.

It is given that the buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates. Therefore, we can say that the buying exchange rates are 95a, 114a, 0.95a.

It is given that the selling exchange rates of each of A, B, and C with respect to L are 10% above their corresponding base exchange rates. Therefore, we can say that the selling exchange rates are 110a, 132a, 1.1a.

We know about the opening and closing units in stock for each currency. Let us draw the table accordingly.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	95a	100a	110a	2500			3300
B	114a	120a	132a	4800			4800
C	0.95a	a	1.1a	48000			51000

Let 'p', 'q' and 'r' be the number of units of currency A, B and C bought by the outlet on that day.

Then, we can say that the outlet sold 'p - 800', 'q' and 'r-3000' units of currency A, B and C respectively.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	95a	100a	110a	2500	p	p - 800	3300
B	114a	120a	132a	4800	q	q	4800
C	0.95a	a	1.1a	48000	r	r - 3000	51000

It is given that the amount of L used by the outlet to buy C equals the amount of L it received by selling C.

$$\Rightarrow 0.95a \cdot r = 1.1a \cdot (r - 3000)$$

$$\Rightarrow 0.15r = 3300$$

$$\Rightarrow r = 22000$$

It is also given that the amounts of L used by the outlet to buy A and B are in the ratio 5:3.

$$\Rightarrow \frac{p \cdot 95a}{q \cdot 114a} = \frac{5}{3}$$

$$\Rightarrow p = 2q$$

Also, the amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

$$\Rightarrow \frac{(p - 800) \cdot 110a}{q \cdot 132a} = \frac{5}{9}$$

$$\Rightarrow \frac{(2q - 800) \cdot 110a}{q \cdot 132a} = \frac{5}{9}$$

$$\Rightarrow q = 600$$

$$\text{Therefore, } p = 2q = 2 \cdot 600 = 1200.$$

It is given that the outlet received 88000 units of L by selling A during the day.

$$\Rightarrow (p - 800) \cdot 110a = 88000$$

$$\Rightarrow (1200 - 800) \cdot 110a = 88000$$

$$\Rightarrow 44000a = 88000$$

$$\Rightarrow a = 2$$

We can fill the entire table and answer all the questions.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	190	200	220	2500	1200	400	3300
B	228	240	264	4800	600	600	4800
C	1.9	2	2.2	48000	22000	19000	51000

From the table we can see that the currency outlet sold 19000 units of currency C. Hence, option B is the correct answer.

 VIDEO SOLUTION

29. A

Let us make note of the information given in the set.

The set states that rural kids were surveyed from 3 regions NE, West and South.

50 students each were surveyed from 150 villages in the NE, 250 villages from the West and 200 villages from the South.

$$\text{Total number of kids from the NE} = 150 \cdot 50 = 7500$$

$$\text{Total number of kids from the West} = 250 \cdot 50 = 12500$$

$$\text{Total number of kids from the South} = 200 \cdot 50 = 10,000.$$

The table, given in the question, gives the number of students whose mothers dropped out before completing primary education. Therefore, the mothers of the remaining students should have completed primary education.

There are 7500 students from the NE in total. The mothers of 5000 of those students dropped out before completing primary education. Therefore, the remaining 2500 kids should have mothers who have completed primary education.

Let us make 2 tables - one representing the number of students and the other representing the number of students whose mother has completed primary education.

55% of the surveyed kids studied in schools run by the Government.

Number of kids studying in Govt. School should be $0.55 \times 30000 = 16,500$

From the given table, we know that the mothers of 13,500 kids who went to Govt. Schools dropped out of primary school.

Therefore, the remaining $16,500 - 13,500 = 3000$ kids should have mother who have completed primary school.

37% of the surveyed kids study in private schools.

Number of kids studying in private schools should be $0.37 \times 30000 = 11,100$.

Number of kids in private schools whose mothers have completed primary school = $11,100 - 2700 = 8400$.

8% of the surveyed kids did not go to school.

Number of kids who did not go to school = $0.08 \times 30000 = 2400$

Number of kids not going to schools whose mothers have completed primary school = $2400 - 1800 = 600$.

In S, 60% of the surveyed kids were in G.

Therefore, 6000 kids in S must be from G.

In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.

There are 300 O kids whose mothers dropped out before completing primary education. These kids represent 50% of the total number of O-kids. Therefore, there must be 600 O-kids from NE - 300 kids should have mothers who dropped out before completing primary education and 300 kids should have mothers who have completed primary education.

The number of kids in G in NE was the same as the number of kids in G in W. Therefore, the number of kids in G in NE and the number of kids in G in W should be equal to 5250. Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250			12500
S	6000			10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050			5200
S	900			4300
Total	3000	8400	600	12000

We have been given that in S, all surveyed kids whose mothers had completed primary education were in school. Therefore, the number of kids not going to school whose mothers have completed primary education in S should be 0. Filling the tables, we get,

Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250	5750	1500	12500
S	6000	3700	300	10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050	3850	300	5200
S	900	3400	0	4300
Total	3000	8400	600	12000

As we can see from the table, 3700 kids out of the 10,000 kids from S are studying in P. 37% of the total number of students from S were studying in P. Therefore, option A is the right answer.

[VIDEO SOLUTION](#)

30.6

If a customer gives 500 rupee notes as her preferred denomination, the number of 500 rupee notes dispensed must be greater than the number of the notes of other denominations dispensed.

If Rs.3500 is dispensed as 500 rupee notes (7 notes), the remaining 1500 rupees should be dispensed using Rs.100 and Rs.200 notes. The minimum number of notes of other denomination required in this case will be 8 ($7 \times 200 + 1 \times 100$). Therefore, at least Rs.4000 should be dispensed as 500 rupee notes.

Case (1):

Rs.4000 is dispensed using 500 rupee notes, 8 five hundred rupee notes will be dispensed.

The remaining 1000 rupees cannot be fully dispensed as 100 rupee notes (since 10 notes will be required).

If 800 rupees is dispensed as 100 rupee notes, then 9 notes will be required to dispense 1000 rupees ($8 \times 100 + 200$).

Therefore, we can eliminate these 2 cases.

If 600 rupees is dispensed using 100 rupee notes, then a minimum of 8 notes will be required to dispense 1000 rupees ($6 \times 100 + 2 \times 200$). Therefore, we can eliminate this case as well.

If 400 rupees is dispensed using 100 rupee notes, then 7 notes will be required ($4 \times 100 + 3 \times 200$). This is a valid case.

If 200 rupees is dispensed using 100 rupee notes, then 6 notes will be required ($2 \times 100 + 4 \times 200$). This is a valid case.

1000 rupees can be dispensed using 5 notes of Rs.200.

Therefore, there are 3 valid cases.

Case (2):

Rs.4500 is dispensed using 500 rupee notes. 9 five hundred rupee notes will be dispensed in this case.

The remaining 500 rupees can be dispensed as 100 rupee notes (5 notes) or a combination of 100 rupee and 200 rupee notes.

$$200 \times a + 100 \times b = 500$$

'a' can take 0, 1, and 2.

Therefore, there are 3 valid cases.

Case (3):

5000 rupees is dispensed using 10 five hundred rupee notes.

There is only 1 valid case.

It has been given that the ATM could serve only 10 customers with a stock of fifty 500 rupee notes. We have to find the maximum number of customers who could have given Rs.500 as their preference.

The least number of 500 rupee notes required to serve a customer who has given Rs.500 as the preference is 8. Using 50 five hundred rupee notes, we can serve $\lceil 50/8 \rceil = 6$ customers. Therefore, 6 is the correct answer.

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31. A

It has been given that the total number of notes dispensed is the smallest possible. Therefore, we have to minimize the number of notes dispensed in each of the 2 cases given.

The least number of notes required to serve a customer who has given 500 rupees as his preference is 10. 50 customers who have given 500 rupee notes as their preference have to be served. We will require $50 \times 10 = 500$ notes for this purpose.

Let us consider the case when a customer has given Rs.100 as his preference.

As we have seen, minimum number of notes will be required when we maximize the number of five hundred rupee notes as much as possible.

If Rs.4000 is dispensed using 500 rupee notes, the remaining 1000 rupees can be dispensed using ten 100 rupee notes. In this case, the number of 100 rupee notes (10) is greater than the number of 500 rupee notes (8). This is a valid case. We have to find if we can reduce the number of notes required any further.

We cannot increase the number of 500 rupee notes to 9 since only 5 hundred rupee notes can be dispensed, violating the condition that the customer has given 100 as his preferred denomination.

If we replace two 100 rupee notes with one 200 rupee note, then the number of 100 rupee notes will become 6. The number of 500 rupee notes (8) exceeds the number of 100 rupee note (6). Therefore, dispensing 4000 rupees using 500 rupee notes and the rest using 100 rupee notes represents the optimum condition.

The minimum number of notes required to serve 1 customer = 8 (five hundred notes) + 10 (hundred notes) = 18
Number of five hundred notes required to serve 50 customers = $8 \times 50 = 400$

Therefore, the total number of notes required = $400 + 500 = 900$.
Therefore, option A is the right answer.

▶ VIDEO SOLUTION

32. 3

We can make the following table from "the total amount of money kept in the three pouches in the first column of the third row is Rs. 4."

If the minimum and maximum value are 1, then the sum of the three pouches in the middle will be Rs 3.

	Column 1	Column 2	Column 3
Row 1			
Row 2		3 (1,1,1)	
Row 3	4 (1,1,2)		

If we calculate the maximum and minimum value possible for each slot in column 1. For the slot, column 1 and row 1, the maximum value possible is 10{2,4,4} while the minimum value possible is 8{2,2,4}.

Similarly, for the slot, column 1 and row 2, the maximum value possible is 13{3,5,5} while the minimum value possible is 11{3,3,5}.

It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. Thus the sum of coins in a row or column must be a multiple of 9.

So, we can iterate that 10,13,4 ...{27} is the only sum possible for the slots of column 1.

We now know two elements of row 2, thus we can iterate from the maximum and the minimum value possible for the slot {column 3, row 2} that 38 is the only value possible for the slot.

We can make the following table:

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)		
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)		

Similarly, we can find the amount for Column 2.

For the slot, column 2 and row 1, the maximum value possible is 22{6,8,8} while the minimum value possible is 20{6,6,8}.

For the slot, column 2 and row 3, the maximum value possible is 5{1,2,3} while the minimum value possible is 4{1,1,2}.

Thus {20,3,4} is the only solution possible.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	

We can similarly make the following table for the last column.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	6 (1,2,3)
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	10 (2,3,5)

Answer 3

[VIDEO SOLUTION](#)

33. D

We want the minimum vote share, that means both the candidates run staid campaigns. And, both the campaigns should be attacking, since we see that if one candidate runs an attacking campaign whereas the other candidate runs an issues campaign, a fair number of voters get transferred to the other candidate's vote share.

This points to us to a scenario where both the campaigns are staid and attacking, which is nothing but the scenario described in the previous question.

If both Ramya and Amiya run staid campaigns, the intensity of each staid campaign is 1
So total number of voters that would vote for them if they focused on issues will be $20 \times (1+1)\% = 40\%$

This means if they had both ran regarding issues, then they would get 20% of votes each.

We are told that both of them run attacking campaigns,

And the rule for mutual attacking campaign is 10% of voters who would have voted for each candidate will not vote.

That means 10% of 20% of each candidate will not vote, now that it is a mutually attacking campaign.

That means, each candidate receives 18% of the votes.

Total votes received is 36%, which is the minimum possible.

[VIDEO SOLUTION](#)

34. E

Since we do not know the exact prices of the share, we cannot determine who gained the most,

[VIEW SOLUTION](#)

35. A

Let us make note of the information given in the set.

The set states that rural kids were surveyed from 3 regions NE, West and South.

50 students each were surveyed from 150 villages in the NE, 250 villages from the West and 200 villages from the South.

Total number of kids from the NE = $150 \times 50 = 7500$

Total number of kids from the West = $250 \times 50 = 12500$

Total number of kids from the South = $200 \times 50 = 10,000$.

The table, given in the question, gives the number of students whose mothers dropped out before completing primary education. Therefore, the mothers of the remaining students should have completed primary education.

There are 7500 students from the NE in total. The mothers of 5000 of those students dropped out before completing primary education. Therefore, the remaining 2500 kids should have mothers who have completed primary education.

Let us make 2 tables - one representing the number of students and the other representing the number of students whose mother has completed primary education.

55% of the surveyed kids studied in schools run by the Government.
Number of kids studying in Govt. School should be $0.55 \times 30000 = 16,500$

From the given table, we know that the mothers of 13,500 kids who went to Govt. Schools dropped out of primary school.
Therefore, the remaining $16,500 - 13,500 = 3000$ kids should have mother who have completed primary school.

37% of the surveyed kids study in private schools.
Number of kids studying in private schools should be $0.37 \times 30000 = 11,100$.
Number of kids in private schools whose mothers have completed primary school = $11,100 - 2700 = 8400$.

8% of the surveyed kids did not go to school.
Number of kids who did not go to school = $0.08 \times 30000 = 2400$
Number of kids not going to schools whose mothers have completed primary school = $2400 - 1800 = 600$.

In S, 60% of the surveyed kids were in G.
Therefore, 6000 kids in S must be from G.

In NE, among the O kids, 50% had mothers who had dropped out before completing primary education. There are 300 O kids whose mothers dropped out before completing primary education. These kids represent 50% of the total number of O-kids. Therefore, there must be 600 O-kids from NE - 300 kids should have mothers who dropped out before completing primary education and 300 kids should have mothers who have completed primary education.

The number of kids in G in NE was the same as the number of kids in G in W. Therefore, the number of kids in G in NE and the number of kids in G in W should be equal to 5250. Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250			12500
S	6000			10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050			5200
S	900			4300
Total	3000	8400	600	12000

We have been given that in S, all surveyed kids whose mothers had completed primary education were in school. Therefore, the number of kids not going to school whose mothers have completed primary education in S should be 0. Filling the tables, we get,

Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250	5750	1500	12500
S	6000	3700	300	10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050	3850	300	5200
S	900	3400	0	4300
Total	3000	8400	600	12000

Number of kids who were out of school in the initial survey = 2400.

The number of kids who were out of school in NE is 600, W is 1500 and S is 300. 25% of the kids from one area, 100% of the kids from one area and 50% of the kids from another area are transferred to G. As a result, the number of kids who were not in school earlier but studying in G now became 50% of 2400 = 1200.

100% of the kids in W could not have been transferred to G (Since the total number of kids transferred to G is 1200 and the number of kids in W alone is 1500).

Let us assume that 100% of the kids from NE transferred (600). In this case, if 25% of the kids from W are transferred (375), 50% of the kids from S (300) will be transferred. The total in this case would be $600 + 375 + 300 = 1275$. Therefore, we can eliminate this case.

If 100% of the kids from NE (600), 25% of the kids from S (75) and 50% of the kids from W (750) are transferred, then the total would have been 1425. Therefore, we can eliminate this case as well.

If 50% of the kids from W (750), 25% of the kids from NE (150) and 100% of the kids from S (300) are transferred, then the total number of kids transferred to G would have been $750 + 150 + 300 = 1200$. Therefore, this must have been the case.

Number of students transferred to G in W = 750.

Number of students present in G in W earlier = 5250

Total = 6000

Therefore, option A is the right answer.

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36. D

Since, we have been given that a person can be reached only by those who are smaller than him. Hence, 1 cannot be reached by anyone. Thus, option D is the correct answer.

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37. C

Let us go by options.

Option A: In first row, 6 is reached by 3, 1 by none, 2 by none, 4 by 3, 3 by none. Hence, ruled out.

Option B: Third row, every individual is approached by other.

Option D: In first row, 6 is reached by 3, 1 by none, 2 by none, 4 by 3, 3 by none. Hence, ruled out.

Hence, Option C is correct.

[VIDEO SOLUTION](#)

38. E

Let the share prices at 10,11,12,1,2,3 be p,q,r,s,t,u

Since Abdul lost all money, $p > u$

From Dane it can be observed, $s+t+u > p+q+r$

$p > r$ and $u > t$

From Emily it can be observed, $r+u > p+s$

Thus we get sequence $p > u > t > q$ and $r > s$.

From these inequalities it can be inferred that the p is the highest

Since $t > q$; option A is definitely false.

Also $r > s$ so option D is definitely false.

 [VIEW SOLUTION](#)

39.7

It has been given that the customer gives 500 rupee notes as her preferred denomination.

Therefore, the number of 500 rupee notes dispensed must be greater than the number of the notes of other denominations dispensed.

If Rs.3500 is dispensed as 500 rupee notes (7 notes), the remaining 1500 rupees should be dispensed using Rs.100 and Rs.200 notes. The minimum number of notes of other denomination required in this case will be 8 ($7 \times 200 + 1 \times 100$). Therefore, at least Rs.4000 should be dispensed as 500 rupee notes.

Case (1):

Rs.4000 is dispensed using 500 rupee notes, 8 five hundred rupee notes will be dispensed.

The remaining 1000 rupees cannot be fully dispensed as 100 rupee notes (since 10 notes will be required).

If 800 rupees is dispensed as 100 rupee notes, then 9 notes will be required to dispense 1000 rupees ($8 \times 100 + 200$).

Therefore, we can eliminate these 2 cases.

If 600 rupees is dispensed using 100 rupee notes, then a minimum of 8 notes will be required to dispense 1000 rupees ($6 \times 100 + 2 \times 200$). Therefore, we can eliminate this case as well.

If 400 rupees is dispensed using 100 rupee notes, then 7 notes will be required ($4 \times 100 + 3 \times 200$). This is a valid case.

If 200 rupees is dispensed using 100 rupee notes, then 6 notes will be required ($2 \times 100 + 4 \times 200$). This is a valid case.

1000 rupees can be dispensed using 5 notes of Rs.200.

Therefore, there are 3 valid cases.

Case (2):

Rs.4500 is dispensed using 500 rupee notes. 9 five hundred rupee notes will be dispensed in this case.

The remaining 500 rupees can be dispensed as 100 rupee notes (5 notes) or a combination of 100 rupee and 200 rupee notes.

$$200a + 100b = 500$$

'a' can take 0, 1, and 2.

Therefore, there are 3 valid cases.

Case (3):

5000 rupees is dispensed using 10 five hundred rupee notes.

There is only 1 valid case.

Total number of valid cases = $3 + 3 + 1 = 7$.

Therefore, 7 is the right answer.

 [VIDEO SOLUTION](#)

40. E

Two ingredients are mixed in equal proportion. So, the minimum amount of protein and carbohydrate should be 60 each, the minimum amount of minerals is 10 and the maximum amount of fat is 50. The combination of O and S satisfies this requirement.

CAT Previous Papers PDF

41. A

The distance traveled by the car in particular direction is as follows : First 10 km in north then 10 km in west then 20 km north then 40 km east then again 10 km north.

So in all total distance traveled is $10+10+20+40+10 = 90$ kms.

VIEW SOLUTION

42. A

Let us make note of the information given in the set.

The set states that rural kids were surveyed from 3 regions NE, West and South.

50 students each were surveyed from 150 villages in the NE, 250 villages from the West and 200 villages from the South.

Total number of kids from the NE = $150 \times 50 = 7500$

Total number of kids from the West = $250 \times 50 = 12500$

Total number of kids from the South = $200 \times 50 = 10,000$.

The table, given in the question, gives the number of students whose mothers dropped out before completing primary education. Therefore, the mothers of the remaining students should have completed primary education.

There are 7500 students from the NE in total. The mothers of 5000 of those students dropped out before completing primary education. Therefore, the remaining 2500 kids should have mothers who have completed primary education.

Let us make 2 tables - one representing the number of students and the other representing the number of students whose mother has completed primary education.

55% of the surveyed kids studied in schools run by the Government.

Number of kids studying in Govt. School should be $0.55 \times 30000 = 16,500$

From the given table, we know that the mothers of 13,500 kids who went to Govt. Schools dropped out of primary school.

Therefore, the remaining $16,500 - 13,500 = 3000$ kids should have mother who have completed primary school.

37% of the surveyed kids study in private schools.

Number of kids studying in private schools should be $0.37 \times 30000 = 11,100$.

Number of kids in private schools whose mothers have completed primary school = $11,100 - 2700 = 8400$.

8% of the surveyed kids did not go to school.

Number of kids who did not go to school = $0.08 \times 30000 = 2400$

Number of kids not going to schools whose mothers have completed primary school = $2400 - 1800 = 600$.

In S, 60% of the surveyed kids were in G.

Therefore, 6000 kids in S must be from G.

In NE, among the 0 kids, 50% had mothers who had dropped out before completing primary education.

There are 300 0 kids whose mothers dropped out before completing primary education. These kids represent

50% of the total number of O-kids. Therefore, there must be 600 O-kids from NE - 300 kids should have mothers who dropped out before completing primary education and 300 kids should have mothers who have completed primary education.

The number of kids in G in NE was the same as the number of kids in G in W. Therefore, the number of kids in G in NE and the number of kids in G in W should be equal to 5250. Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250			12500
S	6000			10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050			5200
S	900			4300
Total	3000	8400	600	12000

We have been given that in S, all surveyed kids whose mothers had completed primary education were in school. Therefore, the number of kids not going to school whose mothers have completed primary education in S should be 0. Filling the tables, we get,

Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250	5750	1500	12500
S	6000	3700	300	10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050	3850	300	5200
S	900	3400	0	4300
Total	3000	8400	600	12000

Number of kids who were out of school in the initial survey = 2400.

The number of kids who were out of school in NE is 600, W is 1500 and S is 300. 25% of the kids from one area, 100% of the kids from one area and 50% of the kids from another area are transferred to G. As a result, the number of kids who were not in school earlier but studying in G now became 50% of 2400 = 1200.

100% of the kids in W could not have been transferred to G (Since the total number of kids transferred to G is 1200 and the number of kids in W alone is 1500).

Let us assume that 100% of the kids from NE transferred (600). In this case, if 25% of the kids from W are transferred (375), 50% of the kids from S (300) will be transferred. The total in this case would be $600 + 375 + 300 = 1275$. Therefore, we can eliminate this case.

If 100% of the kids from NE (600), 25% of the kids from S (75) and 50% of the kids from W (750) are transferred, then the total would have been 1425. Therefore, we can eliminate this case as well.

If 50% of the kids from W (750), 25% of the kids from NE (150) and 100% of the kids from S (300) are transferred, then the total number of kids transferred to G would have been $750 + 150 + 300 = 1200$. Therefore, this must have been the case.

Already, 5100 kids whose mothers had dropped out were in G in S.

After the transfer, 300 more students will be added to the count.

Therefore, 5400 students whose mothers had dropped out will be in G in S.

Total number of kids whose mothers had dropped out in S = 5700

Percentage = $5400/5700 = 94.73\%$.

Therefore, option A is the right answer.

43.20

Let the total number of gold coins with the old woman be '9n'.

Total value of the assets with the old woman = $50 + 3 \times 30 + 70 + 9n = 210 + 9n$.

We know that the assets have been distributed in the ratio 1:2:3.

Therefore, Neeta must have received $35 + 1.5n$ (by value), Seeta must have received $70 + 3n$ and Geeta must have received $105 + 4.5n$.

Further, it has been given that the gold coins distributed were in the ratio 2:3:4.

Therefore, the number of gold coins with Neeta must be '2n', Seeta must be '3n' and Geeta must be '4n'.

Seeta has 3n gold coins. Therefore, the total value of the assets with her must be 70. Seeta could not have inherited all the flats. Therefore, Seeta must have received the house (worth 50 lakh) and 20 lakh from bank deposits.

We know that Geeta did not receive Rs. 30 lakh from the bank deposits. Therefore, Neeta must have received Rs. 30 lakh.

The remaining 5 lakh must be contributed by the gold coins (Since there is no other asset worth 5 lakh).

$$\Rightarrow 5 + 1.5n = 2n$$

$$\Rightarrow 0.5n = 5$$

$$\Rightarrow n = 10$$

The old woman must have had $10 \times 9 = 90$ gold coins.

Total assets = $210 + 90 \times 1 = 300$ lakh

Neeta has received 50 lakh in total, Seeta has received 100 lakh and Geeta has received 150 lakh.

Geeta must have received 90 lakh from 3 flats. Out of the remaining 60 lakh, $4 \times 10 = 40$ lakh has been contributed by the gold coins. Geeta must have received $150 - 90 - 40 = 20$ lakh from bank deposits. Therefore, 20 is the right answer.

44.D

It is given that the base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. Let us assume that base exchange rates are '100a', '120a' and 'a' in that order.

It is given that the buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates. Therefore, we can say that the buying exchange rates are 95a, 114a, 0.95a.

It is given that the selling exchange rates of each of A, B, and C with respect to L are 10% above their corresponding base exchange rates. Therefore, we can say that the selling exchange rates are 110a, 132a, 1.1a.

We know about the opening and closing units in stock for each currency. Let us draw the table accordingly.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	95a	100a	110a	2500			3300
B	114a	120a	132a	4800			4800
C	0.95a	a	1.1a	48000			51000

Let 'p', 'q' and 'r' be the number of units of currency A, B and C bought by the outlet on that day.

Then, we can say that the outlet sold 'p - 800', 'q' and 'r - 3000' units of currency A, B and C respectively.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	95a	100a	110a	2500	p	p - 800	3300
B	114a	120a	132a	4800	q	q	4800
C	0.95a	a	1.1a	48000	r	r - 3000	51000

It is given that the amount of L used by the outlet to buy C equals the amount of L it received by selling C.

$$\Rightarrow 0.95a * r = 1.1a * (r - 3000)$$

$$\Rightarrow 0.15r = 3300$$

$$\Rightarrow r = 22000$$

It is also given that the amounts of L used by the outlet to buy A and B are in the ratio 5:3.

$$\Rightarrow \frac{p * 95a}{q * 114a} = \frac{5}{3}$$

$$\Rightarrow p = 2q$$

Also, the amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

$$\Rightarrow \frac{(p - 800) * 110a}{q * 132a} = \frac{5}{9}$$

$$\Rightarrow \frac{(2q - 800) * 110a}{q * 132a} = \frac{5}{9}$$

$$\Rightarrow q = 600$$

Therefore, $p = 2q = 2 * 600 = 1200$.

It is given that the outlet received 88000 units of L by selling A during the day.

$$\Rightarrow (p - 800) * 110a = 88000$$

$$\Rightarrow (1200 - 800) * 110a = 88000$$

$$\Rightarrow 44000a = 88000$$

$$\Rightarrow a = 2$$

We can fill the entire table and answer all the questions.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	190	200	220	2500	1200	400	3300
B	228	240	264	4800	600	600	4800
C	1.9	2	2.2	48000	22000	19000	51000

From the table we can see that the buying exchange rate of currency C with respect to currency L was 1.9. Hence, we can say that option D is the correct answer.

[▶ VIDEO SOLUTION](#)

45. A

It has been given that the customer has to receive 20 notes at the maximum. Also, we have restriction on the number of 500 rupee notes (fifty) but we do not have any restriction on the number of notes of other denominations. Therefore, in order to serve the maximum number of customers, we have to minimize the number of 500 rupee notes dispensed as much as possible.

If no 500 rupee note is dispensed, then a minimum of 25 notes will be required (25 200 rupee notes).

If one 500 rupee note is dispensed, then a minimum of one 100 rupee note and twenty two 200 rupee notes will be required. The total number of notes required = $1 + 1 + 22 = 24$. Therefore, we can eliminate this case.

If two 500 rupee notes are dispensed, then a minimum of 20 two hundred rupee notes will be required. We can eliminate this case as well since the number of notes required is greater than 20.

If three 500 rupee notes are dispensed, then a minimum of 1 hundred rupee note and 17 two hundred rupee notes will be required. The number of notes required in this case is $3 + 1 + 17 = 21$. Therefore, we can eliminate this case as well.

If four 500 rupee notes are dispensed, then a minimum of 15 two hundred rupee notes will be required. Total number of notes required in this case is $4+15 = 19 < 20$. Therefore, this is a valid case.

The least number of 500 rupee notes with which we can serve a customer such that the total number of notes dispensed does not exceed 20 is 4. Therefore, a maximum of $\lceil 50/4 \rceil = 12$ customers can be served with 50 five hundred rupee notes and hence, option A is the right answer.

 VIDEO SOLUTION



46. B

We are looking for the minimum possible number of votes that Ramya can get and maximise the number of votes that Amiya can get.

We can borrow the scenario from the previous question where Ramya runs an attacking campaign, and we minimised the number of votes she can get.

To minimise the number of votes, we can have Ramya run a staid campaign to minimise the votes, so minimum intensity, which will get her 20% of the votes if she ran with issues. Now that she is running with attacking, she will lose 20% of the votes to Amiya and 5% of the votes will not vote anymore.

That is a total 25% loss. Remaining votes she will get is 75% of the 20% which will leave her with 15% of the votes.

And to maximise the number of votes Amiya can get, we will have her run an vigorous issues campaign, which will give her $2 \times 20\%$ of the votes, that is 40% of the votes. And since Ramya has been running an attacking campaign, 20% of her votes are transferred to Amiya. 20% of the 20% of the votes which is 4% that were going to Ramya will now go to Amiya. That will bring up Amiya's tally up to 44% leaving Ramya's tally at 15%.

The difference in the votes will be $44-15=29\%$.

This is the maximum possible vote difference between the two candidates that is possible.

 VIDEO SOLUTION

47. E

P,Q,S contain 80%,10% and 45% carbohydrates respectively.

Option A,C and D does not contain 60% carbohydrates.

Option E: Carbohydrate % = $\frac{(4 \times 80) + (1 \times 10) + (1 \times 45)}{6} = 62.5\%$

Cost per unit = $\frac{(4 \times 50) + (200) + 100}{6} = 83.3$

Option B: Carbohydrate % = $\frac{(4 \times 80) + (1 \times 20) + (2 \times 45)}{7} = 61.4\%$

Cost per unit = $\frac{(4 \times 50) + (200) + (2 \times 100)}{7} = 600/7 = 85.7$

 VIDEO SOLUTION

48. C

The table represent the platforms and by how many platforms can it be reached. For example, in row 1 and column 1, platform with height 6 can be reached by 3 people.

6(3)	1(0)	2(1)	4(3)	3(0)
9(4)	5(2)	3(2)	2(0)	8(5)
7(1)	8(4)	4(1)	6(6)	5(1)
3(1)	9(4)	5(3)	1(0)	2(1)
1(0)	7(2)	6(2)	3(1)	9(6)

Hence, there are a total of 7 people.

 VIDEO SOLUTION

49. A

We are given the following table,

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

This table compares the GDP, GDP per capita, and population of eight countries, all of which are referenced using a consistent set of units.

Specifically, the GDP values are all given with the same country as the reference point (The reference point being Country 9)

Similarly, the GDP per capita among the eight countries in the table is given using the same reference point (The reference point being Country 10)

As a result, when comparing two countries from the list in terms of GDP, GDP per capita, or population, we can directly use the values presented in the table, since they are all based on a uniform reference point.

We can compare the population of two countries using the formula, $Population = \frac{GDP}{GDP\ Per\ Capita}$

For example, to compare the GDP of Country A and Country B, we can simply use the values listed under "GDP" in the table, as both are referenced using the same country throughout. This eliminates the need for additional conversions or adjustments.

The given question asks us to compare the population of countries between 1, 4, 5, and 7 in the year 2027

Country 1: Population in 2024 will be $0.15/0.41$ or $15/41$ and the population is decreasing at the rate 0.12%

Country 4: Population in 2024 will be $0.12/0.38$ or $12/38$ and the population is increasing at the rate of 0.49%

Country 5: Population in 2024 will be $0.1/0.36$ or $10/36$ and the population is increasing at the rate 0.31%

Country 7: Population in 2024 will be $0.08/0.3$ or $8/30$ and the population is decreasing at the rate 0.11%

Simplifying and comparing the four fractions in order, $\frac{15}{41}, \frac{6}{19}, \frac{5}{18}, \frac{4}{15}$

The decimal values of the following are: 0.36586, 0.315789, 0.2777, 0.2666

Right away we can eliminate Country 7 since the population is not only the least, but it is decreasing, so there is no chance it will have the highest population in three years time.

For the other three countries,

Country 1: $(0.36585) \times (0.9988)^3$ this equals 0.36453

Country 4: $(0.315789) \times (1.0049)^3$ this equals 0.32045

Country 7: $(0.2777) \times (1.0031)^3$ this equals 0.2803612

Hence, Country 1 will have the largest population among these four countries.

50. D

Option A: P and Q in 4:1 ratio gives fat 10% but the protein is 22%

Option B: P and S do not contain fat. Hence we won't analyse the option.

Option C: P and R in 3:1 ratio gives fat 10% but the protein is 27.5%

Option D: Q and S in 1:4 gives fat and protein as 10% and 46 %.

Cost of the mixture = $\frac{[(1 \times 200) + (4 \times 100)]}{5} = 120$

Option E: R and S in 1:3 gives 10% fat and 50% protein.

Cost of the mixture = $\frac{[(1 \times 500) + (3 \times 100)]}{4} = 200$

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51. 1200

It is given that the base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. Let us assume that base exchange rates are '100a', '120a' and 'a' in that order.

It is given that the buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates. Therefore, we can say that the buying exchange rates are 95a, 114a, 0.95a.

It is given that the selling exchange rates of each of A, B, and C with respect to L are 10% above their corresponding base exchange rates. Therefore, we can say that the selling exchange rates are 110a, 132a, 1.1a.

We know about the opening and closing units in stock for each currency. Let us draw the table accordingly.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	95a	100a	110a	2500			3300
B	114a	120a	132a	4800			4800
C	0.95a	a	1.1a	48000			51000

Let 'p', 'q' and 'r' be the number of units of currency A, B and C bought by the outlet on that day.

Then, we can say that the outlet sold 'p - 800', 'q' and 'r - 3000' units of currency A, B and C respectively.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	95a	100a	110a	2500	p	p - 800	3300
B	114a	120a	132a	4800	q	q	4800
C	0.95a	a	1.1a	48000	r	r - 3000	51000

It is given that the amount of L used by the outlet to buy C equals the amount of L it received by selling C.

$$\Rightarrow 0.95a \cdot r = 1.1a \cdot (r - 3000)$$

$$\Rightarrow 0.15r = 3300$$

$$\Rightarrow r = 22000$$

It is also given that the amounts of L used by the outlet to buy A and B are in the ratio 5:3.

$$\Rightarrow \frac{p \cdot 95a}{q \cdot 114a} = \frac{5}{3}$$

$$\Rightarrow p = 2q$$

Also, the amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

$$\Rightarrow \frac{(p - 800) * 110a}{q * 132a} = \frac{5}{9}$$

$$\Rightarrow \frac{(2q - 800) * 110a}{q * 132a} = \frac{5}{9}$$

$$\Rightarrow q = 600$$

$$\text{Therefore, } p = 2q = 2 * 600 = 1200.$$

It is given that the outlet received 88000 units of L by selling A during the day.

$$\Rightarrow (p - 800) * 110a = 88000$$

$$\Rightarrow (1200 - 800) * 110a = 88000$$

$$\Rightarrow 44000a = 88000$$

$$\Rightarrow a = 2$$

We can fill the entire table and answer all the questions.

Currency	Buying rate	Base rate	Selling rate	No. of units			
				Opening stock	Buy	Sell	Closing stock
A	190	200	220	2500	1200	400	3300
B	228	240	264	4800	600	600	4800
C	1.9	2	2.2	48000	22000	19000	51000

From the table we can see that the currency outlet bought 1200 units of A.

[VIDEO SOLUTION](#)

52. C

Let '100x' be the number of smartphones sold in year 2016.

Total revenue generated by Azra = $40x * 15000 = \text{Rs. } 600000x$

Profitability is defined as the profit as a percentage of the revenue. Therefore, profit generated by Azra = $\frac{10}{100} * 600000x = \text{Rs. } 60000x$

Total revenue generated by Bysi = $25x * 20000 = \text{Rs. } 500000x$

Profit generated by Bysi = $\frac{30}{100} * 500000x = \text{Rs. } 150000x$

Total revenue generated by Cxqi = $15x * 30000 = \text{Rs. } 450000x$

Profit generated by Cxqi = $\frac{40}{100} * 450000x = \text{Rs. } 180000x$

Total revenue generated by Dipq = $20x * 25000 = \text{Rs. } 500000x$

Profit generated by Dipq = $\frac{30}{100} * 500000x = \text{Rs. } 150000x$

We can see that profit generated by Cxqi is the highest among all four brands. Hence, option C is the correct answer.

[VIDEO SOLUTION](#)

53. A

Let '100x' be the number of smartphones sold in year 2016.

Total revenue generated by Azra = $40x * 15000 = \text{Rs. } 600000x$

Profitability is defined as the profit as a percentage of the revenue. Therefore, profit generated by Azra = $\frac{10}{100}$
 $*600000x = \text{Rs. } 600000x$

Total revenue generated by Bysi = $25x * 20000 = \text{Rs. } 500000x$

Profit generated by Bysi = $\frac{30}{100} * 500000x = \text{Rs. } 150000x$

Total revenue generated by Cxqi = $15x * 30000 = \text{Rs. } 450000x$

Profit generated by Cxqi = $\frac{40}{100} * 450000x = \text{Rs. } 180000x$

Total revenue generated by Dipq = $20x * 25000 = \text{Rs. } 500000x$

Profit generated by Dipq = $\frac{30}{100} * 500000x = \text{Rs. } 150000x$

It is given that the market sales increased by 40%. Therefore, the number of smartphones sold in 2017 = $1.4 * 100x = 140x$

It is given that Cxqi offered a 40% discount on its unit selling price in 2017 i.e. selling price in 2017 = $0.6 * 30000 = \text{Rs. } 18000$

Also Cxqi's market share increased by 15% whereas the other three brands lost 5% market share.

Brand	Market Share (%)	No of units	Unit Selling Price (Rs.)	Profitability(%)
Azra	35	49x	15,000	10
Bysi	20	28x	20,000	30
Cxqi	30	42x	18,000	20
Dipq	15	21x	25,000	30

Amount of profit generated by Azra = $\frac{10}{100} * 15000 * 49x = 73500x$

Amount of profit generated by Bysi = $\frac{30}{100} * 20000 * 28x = 168000x$

Amount of profit generated by Cxqi = $\frac{20}{100} * 18000 * 42x = 151200x$

Amount of profit generated by Dipq = $\frac{30}{100} * 25000 * 21x = 157500x$

We can see that profit of brands Azra, Bysi and Dipq increased in the year 2017 as compared to 2016. Hence, option A is the correct answer.

 VIDEO SOLUTION

54. B

Let the total number of gold coins with the old woman be '9n'.

Total value of the assets with the old woman = $50 + 3 * 30 + 70 + 9n = 210 + 9n$.

We know that the assets have been distributed in the ratio 1:2:3.

Therefore, Neeta must have received $35 + 1.5n$ (by value), Seeta must have received $70 + 3n$ and Geeta must have received $105 + 4.5n$.

Further, it has been given that the gold coins distributed were in the ratio 2:3:4.

Therefore, the number of gold coins with Neeta must be '2n', Seeta must be '3n' and Geeta must be '4n'.

Seeta has '3n' gold coins. Therefore, the total value of the assets with her must be 70. Seeta could not have inherited all the flats. Therefore, Seeta must have received the house (worth 50 lakh) and 20 lakh from bank deposits.

We know that Geeta did not receive Rs. 30 lakh from the bank deposits. Therefore, Neeta must have received Rs. 30 lakh.

The remaining 5 lakh must be contributed by the gold coins (Since there is no other asset worth 5 lakh).

$\Rightarrow 5 + 1.5n = 2n$

$$\Rightarrow 0.5n = 5$$

$$\Rightarrow n = 10$$

The old-woman must have had $10 \times 9 = 90$ gold coins. Therefore, option B is the right answer.

 VIDEO SOLUTION